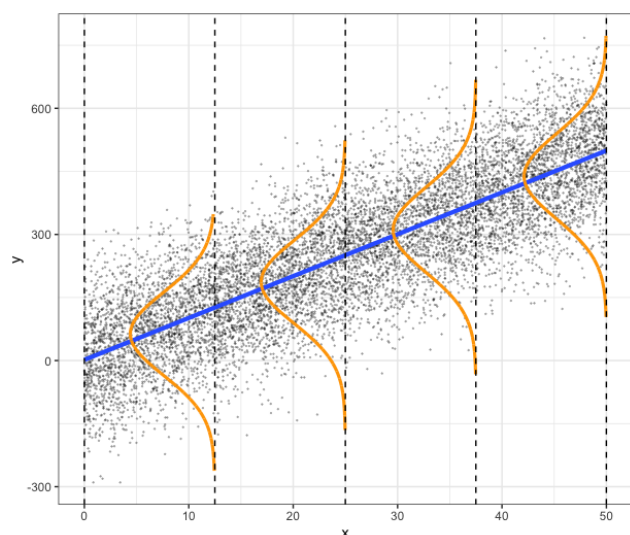


Applied Statistical Inference

Section 1

The Linear Model

1.1 Set up



Definition 1.1. For all models of interest, we have observations of the form $(\mathbf{x}, Y)$, where the entries of the row vector $\mathbf{x} \in \mathbf{R}^{1 \times p}$ are known as **predictors** or **covariates** in different fields and $y \in \mathbf{R}$ is the **response variable**.

Remark 1.2. While $\mathbf{x}$ and Y can be considered as random variables, we will take the covariates to be known, i.e. we will formulate models in terms of conditional distributions of the form $Y|\mathbf{x}$.

Figure 1.1: The linear model assumes a linear relationship between $E[Y|\mathbf{x}]$ and any covariates $\mathbf{x}$, and normally distributed errors with constant variance.

Constant variance linear model

Definition 1.3. The **systematic component** of the linear model specifies the mean of the response variable, as a function of the covariates $\mathbf{x}$ and a vector $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p) \in \mathbf{R}^{p \times 1}$ of fixed parameters,

$$\mu_{\mathbf{x}} = \mathbb{E}[Y|\mathbf{x}] = \sum_{j=1}^p x_j \beta_j = \mathbf{x}\boldsymbol{\beta}.$$

Definition 1.4. The **random component** of a linear model can be specified **parametrically**, by giving the full distribution of $Y|\mathbf{x}$, or **semi-parametrically** by specifying just the variance $\sigma_{\mathbf{x}}^2 = \text{Var}[Y|\mathbf{x}]$. The most common setting is to specify the model parametrically as $Y|\mathbf{x} \sim N(\mu_{\mathbf{x}}, \sigma^2)$, where σ^2 is constant, independent of $\mathbf{x}$.

Remark 1.5. The aim is to determine reasonable estimates for the vector of parameters $\boldsymbol{\beta}$ based on a random sample $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, together with a measure of uncertainty in the estimates.

1.2 Parameter estimates

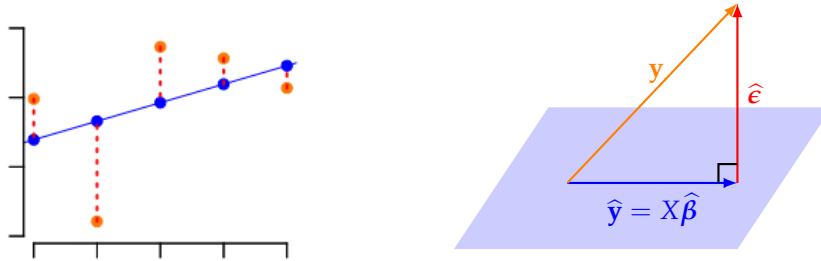


Figure 1.2: The linear model decomposes the vector $\mathbf{y}$ into the component $\hat{\mathbf{y}}$ explained by the columns of $\mathbf{X}$ and the component $\hat{\boldsymbol{\epsilon}}$ orthogonal to the columns of $\mathbf{X}$.

Maximum likelihood estimation

Taking the response variable $\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$, we write down the likelihood function for the parameters $\boldsymbol{\beta}, \sigma^2$, which is just the probability density function for an observed sample $\mathbf{y}$, considered as a function of the parameters.

$$L(\boldsymbol{\beta}, \sigma^2; \mathbf{y}) = \frac{1}{(2\pi\sigma^2)^{\frac{n}{2}}} \exp\left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right).$$

The log-likelihood is

$$l(\boldsymbol{\beta}, \sigma^2; \mathbf{y}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Estimation of β

The way in which the log likelihood depends on the parameters β and σ^2 means that we can estimate these quantities separately.

The gradient of the log likelihood with respect to β is

$$\nabla_{\beta} l(\beta, \sigma^2; \mathbf{y}) = \frac{1}{\sigma^2} X^T (\mathbf{y} - X\beta).$$

Setting the gradient to zero for a maximum, we see that the maximum likelihood $\hat{\beta}$ satisfies

$$X^T \mathbf{y} = X^T X \hat{\beta}.$$

So long as $X^T X$ is invertible, which is the case if X is of full rank, we have

$$\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}.$$

Remark 1.6. The zero-gradient condition for $\hat{\beta}$ has a geometric interpretation shown in Figure 1.2: the vector $\mathbf{e} = \mathbf{y} - X\hat{\beta}$ of residuals is orthogonal to each column of the design matrix.

Properties of the MLE $\hat{\beta}$

We have seen that $\hat{\beta} = (X^T X)^{-1} X^T \mathbf{Y}$. The following properties quickly follow.

1. $\hat{\beta}$ is a **linear estimator**: it is a linear transformation of the vector $\mathbf{Y}$.
2. As a linear transformation of a multivariate normal vector, $\hat{\beta}$ must also be multivariate normal.
3. By linearity of expectation, we see that $\hat{\beta}$ is unbiased for β ,

$$\mathbb{E}[\hat{\beta}] = (X^T X)^{-1} X^T \mathbb{E}[\mathbf{Y}] = (X^T X)^{-1} X^T X \beta = \beta.$$

4. Using bilinearity of the variance operator, the sampling variance of $\hat{\beta}$ is,

$$\text{Var}[\hat{\beta}] = (X^T X)^{-1} X^T \text{Var}[\mathbf{Y}] \left((X^T X)^{-1} X^T \right)^T = (X^T X)^{-1} X^T \sigma^2 I_n X (X^T X)^{-1} = \sigma^2 (X^T X)^{-1}.$$

5. **Gauss-Markov theorem.** Amongst all linear unbiased estimators of β , the least squares estimator $\hat{\beta}$ has the lowest variance. (Proof omitted.)

Estimating σ^2

The gradient of the log-likelihood with respect to σ^2 is

$$\nabla_{\sigma^2} l(\boldsymbol{\beta}, \sigma^2; \mathbf{y}) = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

To maximize the likelihood, we set $\nabla_{\sigma^2} l(\boldsymbol{\beta}, \sigma^2; \mathbf{y}) = 0 = \nabla_{\boldsymbol{\beta}} l(\boldsymbol{\beta}, \sigma^2; \mathbf{y})$, to get

$$\hat{\sigma}^2 = \frac{(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})}{n}.$$

Remark 1.7. It is straightforward to check that the stationary points we have found are indeed maxima by considering the hessian $\nabla^2 l$.

Remark 1.8. Note that $\hat{\sigma}^2$ is biased downwards for σ^2 . More commonly, we instead use the unbiased estimator S^2 in which the denominator $n - r$ replaces n . This will be shown to be unbiased on Problem Sheet 1.

1.3 Fitted values and residuals

Definition 1.9. For a linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, the fitted values are given by $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$.

Remark 1.10. From the definition of $\hat{\boldsymbol{\beta}}$, the fitted values are clearly a linear function of $\mathbf{Y}$,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} = \mathbf{P}\mathbf{Y}.$$

The matrix $\mathbf{P}$ is often called the **hat matrix**, since it puts a hat on $\mathbf{Y}$.

Theorem 1.11. The matrix $\mathbf{P}$ is the orthogonal projection onto the column space of the design matrix $\mathbf{X}$.

Proof. There are two properties to be shown,

1. $\mathbf{P}$ is symmetric, i.e. $\mathbf{P}^T = \mathbf{P}$.
2. $\mathbf{P}$ is idempotent, i.e. $\mathbf{P}^2 = \mathbf{P}$.

This is an exercise on Problem Sheet 1.

Remark 1.12. It can be shown that $\mathbf{P}$ has r eigenvalues equal to 1 and $n - r$ eigenvalues equal to zero, where $r = \text{rank}(\mathbf{X})$.

Proposition 1.13. The vector of fitted values has a multivariate normal distribution $\hat{\mathbf{Y}} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{P})$. Note that this multivariate normal distribution is degenerate: although $\hat{\mathbf{Y}} \in \mathbf{R}^n$, its probability density function is only non-zero on an r -dimensional subspace, namely the column space of $\mathbf{X}$.

Proposition 1.14. For any two observations i and j , the covariance between the two fitted values is given by

$$\begin{aligned}\text{Cov}(\hat{Y}_i, \hat{Y}_j) &= \text{Cov}\left(\sum_k P_{ik} Y_k, \sum_l P_{jl} Y_l\right) = \sum_k \sum_l P_{ik} P_{jl} \text{Cov}(Y_k, Y_l) \\ &= \sum_k \sigma^2 P_{ik} P_{jk} = \sum_k \sigma^2 P_{ik} P_{kj} = \sigma^2 (P^2)_{ij} = \sigma^2 P_{ij},\end{aligned}$$

where we have freely used the bilinearity of $\text{Cov}(\cdot, \cdot)$ and the independence of the entries of $\mathbf{Y}$.

Definition 1.15. The vector of residuals is $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I} - \mathbf{P})\mathbf{Y}$.

Proposition 1.16. The vector of residuals has a multivariate normal distribution, $\mathbf{e} \sim N(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{P}))$. As with the fitted values, the joint distribution of the residuals is degenerate multivariate normal, with probability concentrated on the $n - r$ -dimensional subspace orthogonal to the column space of $\mathbf{X}$.

Proposition 1.17. The covariance between any two residuals is given by

$$\text{Cov}(e_i, e_j) = \sigma^2(\mathbf{I} - \mathbf{P})_{ij},$$

by essentially the same argument as above.

Remark 1.18. The matrix $\mathbf{I} - \mathbf{P}$ is the corresponding projection onto the subspace of $\mathbf{R}^n$ orthogonal to the column space of the design matrix.

1.4 Categorical covariates

Binary covariates

Example 1.19. Consider an experiment with two distinct conditions. E.g. a clinical trial, where group 0 receives a placebo and group 1 receives an experimental drug.

A suitable linear model would be,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

where

$$x_i = \begin{cases} 1 & \text{subject } i \text{ in treatment group} \\ 0 & \text{subject } i \text{ in placebo group.} \end{cases}$$

In the placebo group ,

$$E[Y|x = 0] = \beta_0,$$

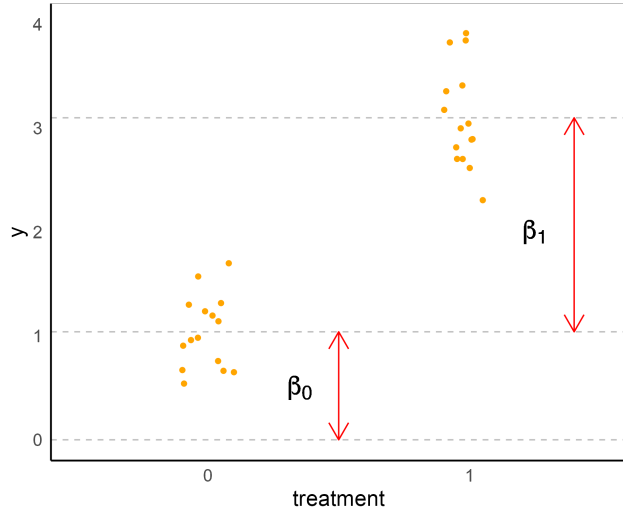
whereas in the treatment group

$$E[Y|x=1] = \beta_0 + \beta_1.$$

In this case, the question of interest is whether or not the treatment effect

$$E[Y|x=1] - E[Y|x=0] = \beta_1$$

is zero.



Example 1.20. Extending the example above, suppose we allow for a difference in mean response according to whether or not the subject has a standard diet or an experimental diet.

$$z_i = \begin{cases} 1 & \text{subject } i \text{ in experimental diet group} \\ 0 & \text{subject } i \text{ in standard diet group} \end{cases}$$

Definition 1.21. The two-way additive model for the binary covariates x and z is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2).$$

Definition 1.22. The two-way interaction model for the binary covariates x and z is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2).$$

Remark 1.23. Figure 1.4 shows the difference between the two-way additive model and the two-way interaction model. The additive model assumes a common treatment effect β_1 across the two diet conditions,

Figure 1.3: Independent samples from two treatment groups. β_0 estimates the mean of group 0 and $\beta_0 + \beta_1$ estimates the mean in group 1.

$$\begin{aligned}\beta_1 &= E[Y|x = 1, z = 0] - E[Y|x = 0, z = 0] \\ &= E[Y|x = 1, z = 1] - E[Y|x = 0, z = 1].\end{aligned}$$

The interaction model allows for the treatment effects to be different in the two diet conditions. Firstly, in the standard diet condition,

$$E[Y|x = 0, z = 0] = \beta_0; \quad E[Y|x = 1, z = 0] = \beta_0 + \beta_1,$$

so that the treatment effect is β_1 . Now, in the experimental condition,

$$E[Y|x = 0, z = 1] = \beta_0 + \beta_2; \quad E[Y|x = 1, z = 1] = \beta_0 + \beta_1 + \beta_2 + \beta_3,$$

so that here the treatment effect is $\beta_1 + \beta_3$.

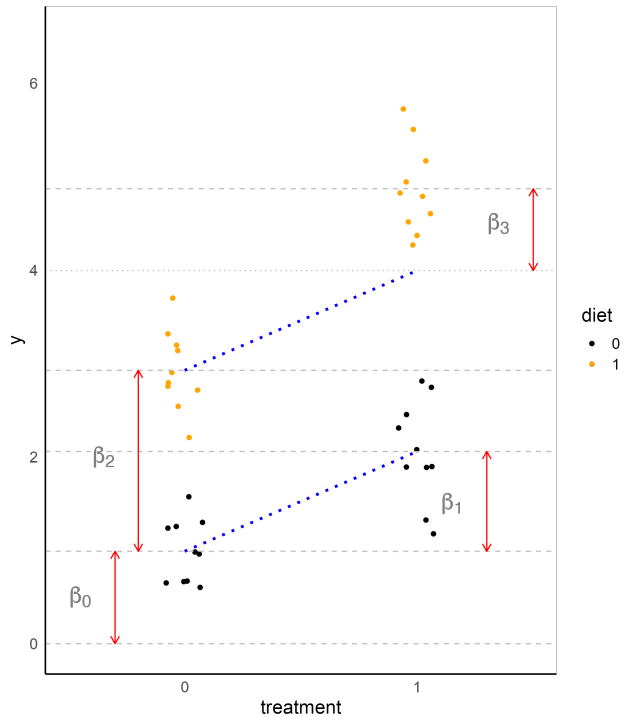


Figure 1.4: Two-way interaction model for the binary covariates *treatment* and *diet*. The blue parallel lines represent the additive treatment effect β_1 . The interaction effect is the additional treatment effect β_3 in diet 1 over and above the treatment effect in diet condition 0.

A single categorical covariate with more than two categories - ANOVA

We have seen that a binary categorical variable can easily be encoded as a single column of the design matrix with entries 0 or 1. In general, if there are k different categories (sometimes called 'levels'), $k - 1$ columns are needed to encode the variable in a model with an intercept. There are different ways of encoding a categorical variable, depending on the comparisons that are of interest in a particular analysis.

Example 1.24. Suppose two different treatments are to be compared with a control condition, with each subject receiving a single treatment. The most general linear model is

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

where for $j \in \{1, 2\}$,

$$x_{ij} = \begin{cases} 1 & \text{subject } i \text{ receives treatment } j \\ 0 & \text{otherwise} . \end{cases}$$

The effect of treatment 1 is

$$E[Y|x_1 = 1] - E[Y|x_1 = 0] = (\beta_0 + \beta_1) - \beta_0 = \beta_1,$$

and similarly for treatment 2,

$$E[Y|x_2 = 1] - E[Y|x_2 = 0] = (\beta_0 + \beta_2) - \beta_0 = \beta_2.$$

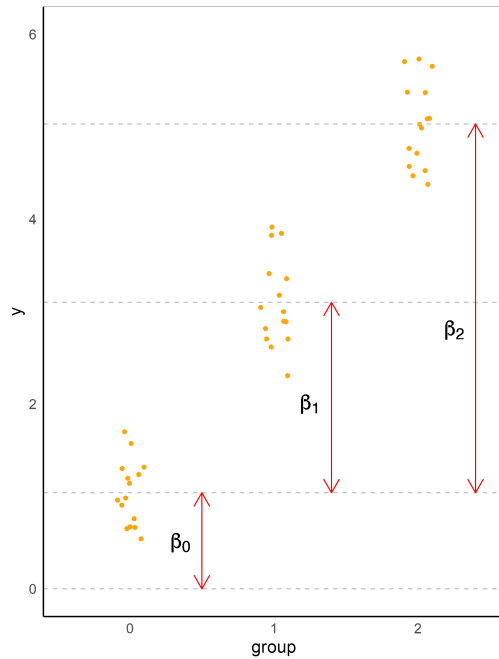


Figure 1.5: Independent samples from three groups with treatment contrasts. The model is parameterized to allow easy comparison of group 1 with group 0 and group 2 with group 0.

Coefficient of determination, R^2 .

Definition 1.25. For the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, we define

- the Total Sum of Squares, $TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$,
- the Residual Sum of Squares $RSS = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n \hat{\epsilon}_i^2$.

Remark 1.26. TSS is just the residual sum of squares for the intercept-only model.

Definition 1.27. The coefficient of determination is

$$R^2 = 1 - \frac{RSS}{TSS}.$$

Remark 1.28. R^2 is a crude measure of goodness of fit. In Figure 1.7, we see four datasets with the same regression line and R^2 , but very different properties. In the presence of individual influential observations or systematic nonlinearities, R^2 can be misleading.

Remark 1.29. If two nested models are fit to the same response variable $\mathbf{Y}$, say

$$\mathbf{Y} = \mathbf{X}_0\boldsymbol{\beta}_0 + \boldsymbol{\epsilon}_0, \quad \text{and} \quad \mathbf{Y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\epsilon}_1,$$

where $\text{span}(\mathbf{X}_0) \subset \text{span}(\mathbf{X}_1)$, then $R_0^2 < R_1^2$: in terms of this measure, bigger models always fit better.

Definition 1.30. The adjusted R^2 is

$$\bar{R}^2 = 1 - \frac{RSS/(n-p)}{TSS/(n-1)}.$$

Remark 1.31. In the adjusted R^2 , the RSS and TSS are adjusted by their respective degrees of freedom. In contrast with R^2 , fitting a bigger model can reduce $\bar{R}^2$.

1.5 Inference

Statistical inference is the process of making statements about the population from which our random sample is taken. This involves quantifying the uncertainty due to sampling, by making assumptions about the probability distributions that describe population-level variability.

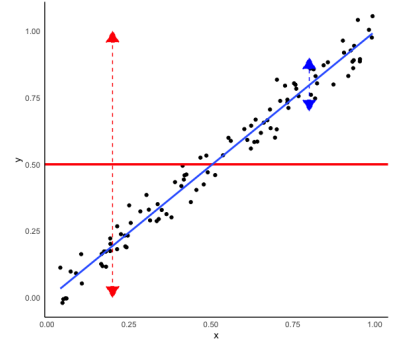


Figure 1.6: When x is not known, the best predictor of y is just the sample mean $\bar{y}$ (red solid line). When x is known, the best predictor is the regression line (blue solid line). The arrows denote the variability around these two predictions. R^2 is a measure of the improvement in precision from using the regression line rather than just the sample mean.

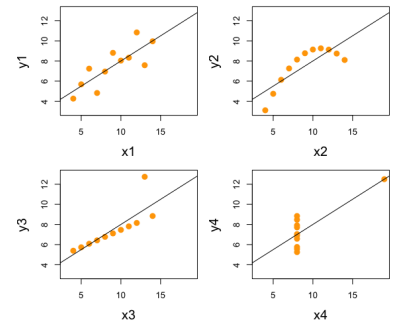


Figure 1.7: Anscombe's quartet: all four data sets share the same regression line and R^2 . This shows that R^2 is a crude summary of goodness of fit.

Confidence intervals

For the normal linear model, we know that

$$V = \frac{RSS}{\sigma^2} \sim \chi_{n-r}^2,$$

where $r = \text{rank}(X)$ and χ_k^2 is the chi-square distribution with k degrees of freedom.

Suppose we are interested in a linear combination of the parameters,

$$\mathbf{c}^T \boldsymbol{\beta} = \sum_{j=1}^p c_j \beta_j.$$

Since $\widehat{\boldsymbol{\beta}}$ is multivariate normal, we have, assuming X is full rank,

$$\mathbf{c}^T \widehat{\boldsymbol{\beta}} \sim N \left(\mathbf{c}^T \boldsymbol{\beta}, \sigma^2 \mathbf{c}^T (X^T X)^{-1} \mathbf{c} \right),$$

so that

$$Z = \frac{\mathbf{c}^T \widehat{\boldsymbol{\beta}} - \mathbf{c}^T \boldsymbol{\beta}}{\sqrt{\sigma^2 \mathbf{c}^T (X^T X)^{-1} \mathbf{c}}} \sim N(0, 1).$$

The random variables $\widehat{\boldsymbol{\beta}}$ and RSS are independent, so that by the definition of the t -distribution,

$$\frac{Z}{\sqrt{V/(n-p)}} \sim t_{n-p},$$

which does not depend on any unknown parameters. This is a **pivot**, which can be manipulated to give a $(1 - \alpha)$ confidence interval for $\mathbf{c}^T \boldsymbol{\beta}$,

$$\left(\mathbf{c}^T \widehat{\boldsymbol{\beta}} - t_{n-p}^{(\alpha/2)} \sqrt{\widehat{\sigma}^2 \mathbf{c}^T (X^T X)^{-1} \mathbf{c}}, \mathbf{c}^T \widehat{\boldsymbol{\beta}} + t_{n-p}^{(\alpha/2)} \sqrt{\widehat{\sigma}^2 \mathbf{c}^T (X^T X)^{-1} \mathbf{c}} \right),$$

where $t_{n-p}^{(\alpha/2)}$ is the appropriate quantile value such that

$$\Pr \left(-t_{n-p}^{(\alpha/2)} < T < t_{n-p}^{(\alpha/2)} \right) = 1 - \alpha, \quad T \sim t_{n-p}.$$

Remark 1.32. This is a random interval, because $\widehat{\boldsymbol{\beta}}$ and $\widehat{\sigma}^2$ are random. It contains the true value $\mathbf{c}^T \boldsymbol{\beta}$ with probability $1 - \alpha$.

Remark 1.33. By selecting $\mathbf{c}$ appropriately, we can use construct confidence intervals for components of $\boldsymbol{\beta}$ e.g. β_1 or contrasts of interest such as $\beta_1 - \beta_2$.

Example 1.34. Consider a designed experiment in which two different treatments are compared against a control. Write the i th observation as

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2),$$

where

$$x_{ij} = \begin{cases} 1 & \text{subject } i \text{ receives treatment } j \quad j \in \{1, 2\} \\ 0 & \text{otherwise} \end{cases}$$

The mean value of the response for subjects receiving treatment 1 is $\beta_0 + \beta_1$, and for those receiving treatment 2 is $\beta_0 + \beta_2$. To obtain a confidence interval for the difference in effectiveness of the two treatments, $\beta_1 - \beta_2$, we apply the result above with $c^T = (0, 1, -1)$.

Example 1.35. (Confidence interval for the population mean) The simplest linear model is the intercept-only model,

$$y_i = \beta_0 + \epsilon_i,$$

in which the observations are all identically distributed.

Here we have

$$X = \begin{pmatrix} \uparrow \\ 1 \\ \downarrow \end{pmatrix},$$

so that $X^T X$ is simply the 1×1 matrix (n) and $X^T \mathbf{y} = \sum_{i=1}^n y_i$, and so

$$\widehat{\beta}_0 = (X^T X)^{-1} X^T \mathbf{y} = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y},$$

as expected. The sampling distribution of our estimator is

$$\widehat{\beta}_0 \sim N\left(\beta_0, \frac{\sigma^2}{n}\right).$$

We also know that the sampling distribution of the RSS is

$$\frac{RSS}{\sigma^2} \sim \chi_{n-1}^2.$$

By Cochran's theorem, $\widehat{\beta}_0$ and RSS are independent, so if $\widehat{\sigma}^2 = \frac{RSS}{n-1}$,

$$\frac{\widehat{\beta}_0 - \beta_0}{1/\sqrt{n}} \bigg/ \sqrt{\widehat{\sigma}^2} \sim t_{n-1}.$$

Hence the random interval

$$\left(\widehat{\beta}_0 - t_{n-1}^{(\alpha/2)} \sqrt{\frac{\widehat{\sigma}^2}{n}}, \widehat{\beta}_0 + t_{n-1}^{(\alpha/2)} \sqrt{\frac{\widehat{\sigma}^2}{n}} \right)$$

is a $1 - \alpha$ confidence interval for the population mean β_0 .

Hypothesis testing

Suppose we have fit a model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, and there is interest in testing whether the data are adequately explained by a sub-model with fewer parameters.

Formally, we wish to test

$$H_0 : \mathbf{E}(\mathbf{Y}) \in \text{span}(\mathbf{X}_0) \quad \text{vs} \quad H_1 : \mathbf{E}(\mathbf{Y}) \notin \text{span}(\mathbf{X}_0),$$

for some matrix $\mathbf{X}_0$ with $\text{span}(\mathbf{X}_0) \subset \text{span}(\mathbf{X})$. The span is the linear space spanned by the column vectors of the matrix.

Theorem 1.36. Under $H_0 : \mathbf{E}(\mathbf{Y}) \in \text{span}(\mathbf{X}_0)$,

$$F = \frac{\text{RSS}_0 - \text{RSS}}{\text{RSS}} \frac{n - r}{r - s} \sim F_{r-s, n-r},$$

where $r = \text{rank}(\mathbf{X})$ and $s = \text{rank}(\mathbf{X}_0)$.

Remark 1.37. Even if the data were generated by the smaller model, the bigger model will fit better because it has more free parameters. The F-statistic measures the improvement in model fit when the larger model is used: the numerator term $\text{RSS}_0 - \text{RSS}$ is always positive. Only if there is more improvement than we expect by chance would we reject the null hypothesis. Hence this is always a one-tailed test.

Remark 1.38. Where the larger model has one additional degree of freedom, the F-test is equivalent to a t-test.

Prediction

One common use for linear models is to predict unknown values of the response given new values $\mathbf{x}_*$ of the covariates.

Definition 1.39. The predicted value y_* corresponding to the covariate values $\mathbf{x}_*$ is

$$y_* = \mathbf{x}_*^T \hat{\boldsymbol{\beta}}.$$

Remark 1.40. The predicted value is a point lying on the regression surface. Given the model, we have no reason to say believe the true value of y lies one side of the surface or the other.

Definition 1.41. The $1 - \alpha$ confidence interval for the mean response at $\mathbf{x}_*$ is

$$\hat{y}_* \pm t_{n-p}^{(\alpha/2)} \sqrt{\hat{\sigma}^2 \mathbf{x}_*^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_*}$$

Remark 1.42. This interval takes into account only the uncertainty from estimating β as $\hat{\beta}$.

Definition 1.43. The $1 - \alpha$ confidence interval for an individual response at $\mathbf{x}_*$ is

$$\hat{y}_* \pm t_{n-p}^{(\alpha/2)} \sqrt{\hat{\sigma}^2 (1 + \mathbf{x}_*^T (X^T X)^{-1} \mathbf{x}_*)}$$

Remark 1.44. This interval takes into account not just the uncertainty from estimating β as $\hat{\beta}$, but also the uncertainty in the true value of y , which will deviate from its expectation $\mathbf{x}_*^T \beta$ by some amount $\epsilon \sim N(0, \sigma^2)$.

Design effects: identifiability and precision

Remark 1.45. The sampling variance of $\hat{\beta}_j$ the estimator of the parameter β_j , is given by the j th diagonal entry of $\sigma^2 (X^T X)^{-1}$. Two things affect the precision of our parameter estimates: the error variance and the design matrix X . Generally, we have no control over σ^2 , but at least in cases of a designed experiment, we can choose the values of covariates to minimize the variance of the parameters we are interested in.

Remark 1.46. Figure 1.8 shows a dataset in which it is not possible to estimate the parameter β_1 , because there is no variability in the variable x . Essentially the same effect occurs in multiple regression in a less trivial way, e.g. a study that attempts to measure the separate effects of length of more- and less- dominant arm length on throwing distance. For typical individuals, the lengths of both arms are similar, and so the covariate vectors $\mathbf{x}_L$ and $\mathbf{x}_R$ would be close to collinear. Hence, the estimates of the corresponding parameters would be highly correlated. In the limit where $\mathbf{x}_L = \mathbf{x}_R$ exactly, the design matrix fails to be full rank.

1.6 Residuals

Leverage

Remark 1.47. In the most common linear model, the error random variables are assumed to be independent and identically distributed, $\epsilon_i \sim N(0, \sigma^2)$. It is important to note that this is not true of the residuals $\hat{\epsilon}$: in general they are neither independent nor identically distributed.

Definition 1.48. If P is the hat matrix such that the residuals are $\hat{\epsilon} = (I - P)Y$, then the leverage of the i th residual is p_{ii} , the i th diagonal entry of P . Note that the leverage of a residual depends only on P , and therefore only on the design matrix X ; it does not depend on Y .

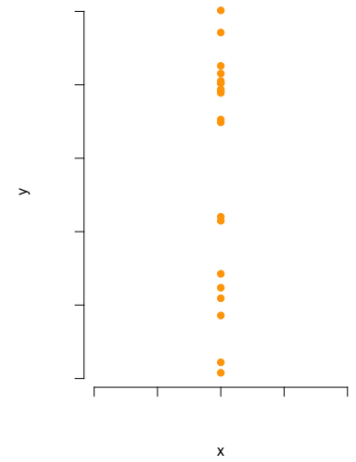


Figure 1.8: For simple regression,

$$y = \beta_0 + \beta_1 x,$$

β_1 is the rate of change of y with x . Clearly we have no information about β_1 if x does not change.

Remark 1.49. $\text{Var}(\hat{e}_i) = \sigma^2(1 - p_{ii})$, so that residuals with high leverage have small variance. Such points can be thought of as pulling the fitted surface towards them.

Example 1.50. In the simple linear model, in which $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, the leverage of residual i is given by

$$\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_k (x_k - \bar{x})^2}.$$

To see this, note that for this model we have

$$X = \begin{pmatrix} \uparrow & \uparrow \\ 1 & x_i \\ \downarrow & \downarrow \end{pmatrix} \quad \text{and} \quad X^T X = \begin{pmatrix} n & \sum_k x_k \\ \sum_k x_k & \sum_k x_k^2 \end{pmatrix},$$

so that

$$\begin{aligned} (X^T X)^{-1} &= \frac{1}{n \sum_k x_k^2 - (\sum_k x_k)^2} \begin{pmatrix} \sum_k x_k^2 & -\sum_k x_k \\ -\sum_k x_k & n \end{pmatrix} \\ &= \frac{1}{n \sum_k (x_k - \bar{x})^2} \begin{pmatrix} \sum_k x_k^2 & -\sum_k x_k \\ -\sum_k x_k & n \end{pmatrix}. \end{aligned}$$

To get the i th diagonal entry of $P = X(X^T X)^{-1} X^T$, we pre- and post-multiply P by the vector $\mathbf{v}_i = (0, \dots, 1, \dots, 0)^T$ with a single 1 in the i th position.

$$\begin{aligned} \mathbf{v}_i^T P \mathbf{v}_i &= \mathbf{v}_i^T X (X^T X)^{-1} X^T \mathbf{v}_i = \begin{pmatrix} 1 & x_i \end{pmatrix} (X^T X)^{-1} \begin{pmatrix} 1 \\ x_i \end{pmatrix} \\ &= \frac{1}{n \sum_k (x_k - \bar{x})^2} \begin{pmatrix} 1 & x_i \end{pmatrix} \begin{pmatrix} \sum_k x_k^2 - x_i \sum_k x_k \\ -\sum_k x_k + n x_i \end{pmatrix} \\ &= \frac{\sum_k x_k^2 - 2x_i \sum_k x_k + n x_i^2}{n \sum_k (x_k - \bar{x})^2} \\ &= \frac{\sum_k (x_k - \bar{x})^2 + n \bar{x}^2 - 2n x_i \bar{x} + n x_i^2}{n \sum_k (x_k - \bar{x})^2} = \frac{\sum_k (x_k - \bar{x})^2 + n (x_i - \bar{x})^2}{n \sum_k (x_k - \bar{x})^2} \\ &= \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_k (x_k - \bar{x})^2}. \end{aligned}$$

Remark 1.51. Since P is a projection matrix, we know that

$$\text{tr}(P) = \sum_{i=1}^n p_{ii} = r = \text{rank}(X),$$

so the average leverage is $\frac{r}{n}$. A common heuristic is to regard an observation as having a large leverage when it is $> \frac{2r}{n}$.

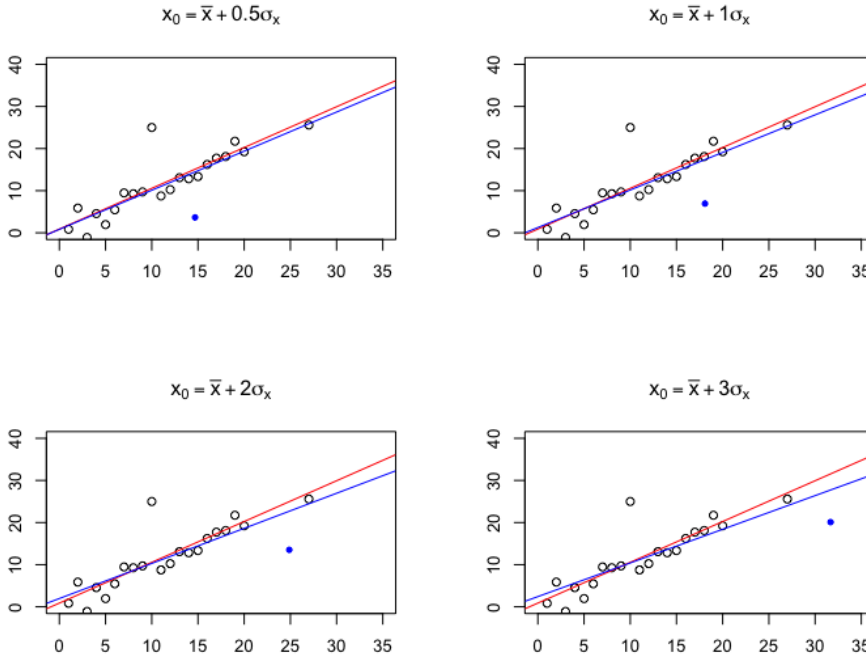


Figure 1.9: The blue and red regression lines respectively are computed with and without blue data point x_0 , whose position relative to $\bar{x}$, the mean of the other data points, is given in the title. As x_0 gets further from $\bar{x}$, its leverage increases and it exerts a greater pull on the regression line.

Standardized residuals

Definition 1.52. The i th standardized residual e_i is given by

$$e_i = \frac{\hat{\epsilon}_i}{\sqrt{\hat{\sigma}^2(1 - p_{ii})}},$$

where $\hat{\sigma}^2$ is the usual unbiased estimate of the variance.

Remark 1.53. Even a single atypical observation can distort the fit of a linear model. It is common to leave out each individual observation in turn when evaluating a model.

Definition 1.54. The i th deleted residual is

$$d_i = y_i - \hat{y}_{(i)},$$

where $\hat{y}_{(i)}$ is the fitted value for observation i when the model is fitted without this observation.

Remark 1.55. It can be shown that

$$d_i = \frac{\hat{\epsilon}_i}{1 - p_{ii}},$$

so that deleted residuals can easily be computed without having to refit the model.

Remark 1.56. When observation i is left out, it can be treated as an unseen observation, so the variance of the i th deleted residual can be estimated as

$$\hat{\sigma}_{(i)}^2 (1 + \mathbf{x}_i^T (\mathbf{X}_{(i)}^T \mathbf{X}_{(i)})^{-1} \mathbf{x}_i),$$

which can be shown to equal

$$\frac{\hat{\sigma}_{(i)}^2}{1 - p_{ii}}.$$

Definition 1.57. the i th studentized deleted residual is

$$t_i = \frac{d_i}{\sqrt{\frac{\hat{\sigma}_{(i)}^2}{1 - p_{ii}}}} = \frac{\hat{\epsilon}_i}{\sqrt{\hat{\sigma}_{(i)}^2 (1 - p_{ii})}};$$

it is sometimes known as the externally studentized residual.

Influence

We have seen above that not all observations exert equal influence on fitted surface of the model. One way to measure an observation's influence is to ask how much the model fit changes when the observation is omitted.

Definition 1.58. The Cook's distance for observation i is given by

$$D_i = \frac{(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})^T \mathbf{X}^T \mathbf{X} (\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})}{r \hat{\sigma}^2} = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{r \hat{\sigma}^2},$$

in which $\hat{\boldsymbol{\beta}}_{(i)}$ is the estimator of $\boldsymbol{\beta}$ made with the i th observation left out and $\hat{y}_{j(i)}$ is the fitted value for observation j when observation i is left out.

Remark 1.59. Cook's distance can be written as a function of the standardized residual e_i , the leverage p_{ii} and the rank r of $\mathbf{X}$ as

$$D_i = \frac{e_i^2 p_{ii}}{r(1 - p_{ii})}.$$

1.7 Residual plots

Residuals versus fitted values

The first residual plot in Figure 1.10 shows the raw (i.e. unstandardized) residuals $\hat{\epsilon}$ against the fitted values $\hat{\mathbf{y}}$. Its purpose is to identify any non-linear relationships that the model has omitted. By default, R adds a local smoother to make it easier to identify patterns. We will see how smoothers work in Section 1.9.

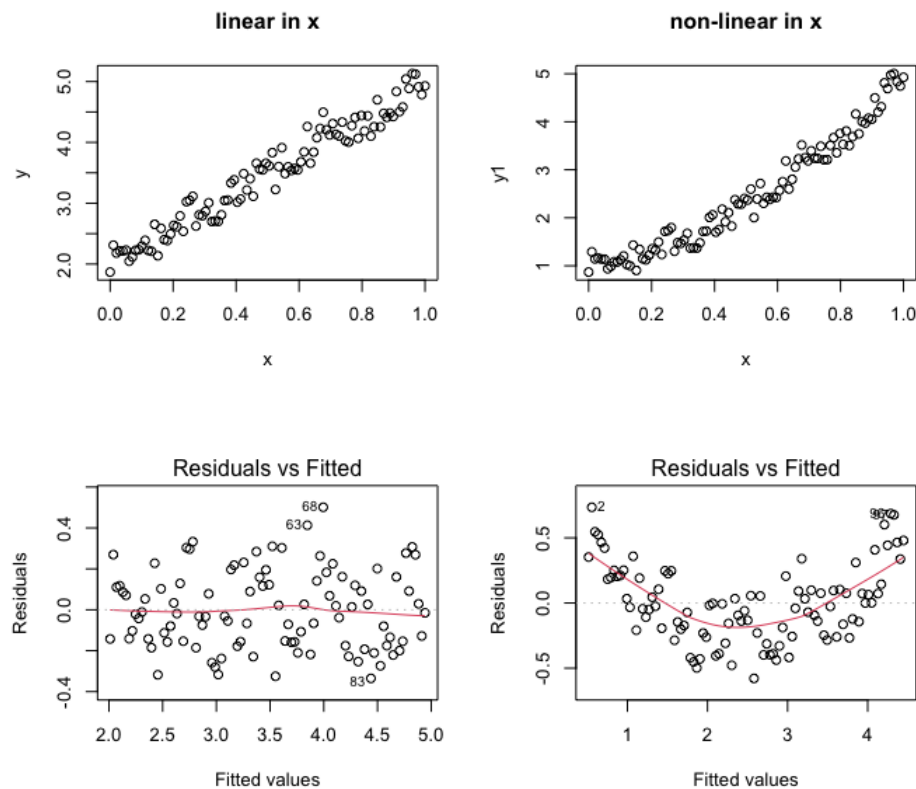


Figure 1.10: The residuals show the non-linear pattern in the response variable more clearly, because the linear effect has been subtracted off. The red line is a local smoother, to guide the eye.

Remark 1.60. We plot the residuals against the fitted values because these are uncorrelated, $\text{Cov}(\hat{\epsilon}, \hat{y}) = 0$. Plotting residuals against the observed data $\mathbf{y}$ would be less helpful, because we know $\hat{\epsilon} = (I - P)\mathbf{y}$, so we would see a linear relationship between these quantities, which could hide problems with the model that we're trying to identify.

Q-Q plot of standardized residuals

Inference in the linear model assumes the errors are normally distributed with constant variance. One way to see if this assumption is reasonable is to see whether the standardized residuals appear to be normally distributed using a quantile-quantile (Q-Q) plot, as shown in Figure 1.11. We review the theory of the Q-Q plot in Section 1.8.

Remark 1.61. Note that standardized or studentized residuals use a plug-in estimate of $\hat{\sigma}^2$ from the data, which means that they will be t distributed rather than normally distributed. For a reasonable large sample, the t distribution is close to normal. In smaller samples, the heavier tails of the t distribution may be evident even when the model is fitting well.

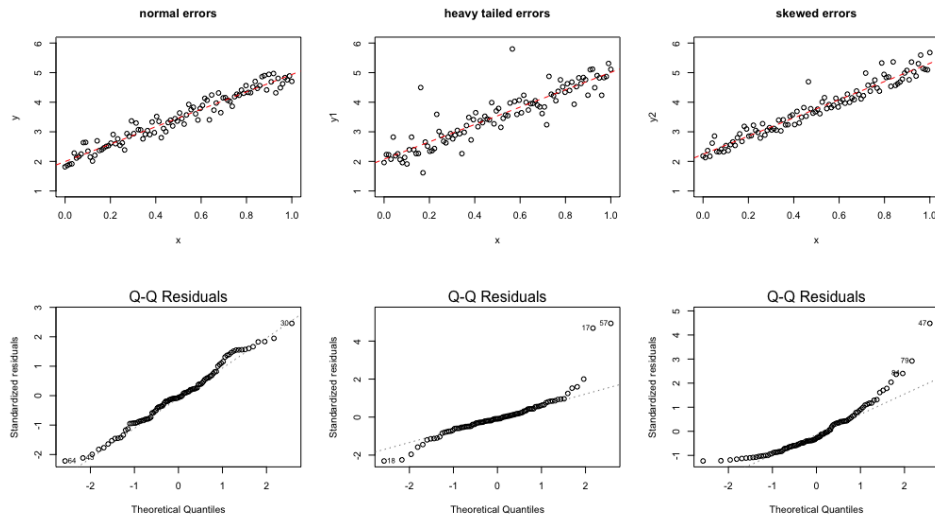


Figure 1.11: The upper left plot shows a linear model with normal errors, and the corresponding Q-Q plot of standardized residuals below: the ordered residuals fall on the identity line. The upper central plot shows a linear model with t -distributed errors, with a heavier tail than the normal. The corresponding Q-Q plot shows a number of large standardized residuals in each tail, and the plot crosses the identity line. The upper right plot shows a linear model with Gamma-distributed errors with a pronounced skew. The right Q-Q plot shows a characteristic bowing pattern, remaining above the identity line.

Scale-location plot

The third residual plot, Figure 1.12 examines the assumption of constant variance. We plot the square root of the absolute value of the standardized residuals against the fitted values.

Remark 1.62. We plot the absolute value so that it is easier to see the magnitude of the spread. We plot the standardized residuals to account for differences in leverage.

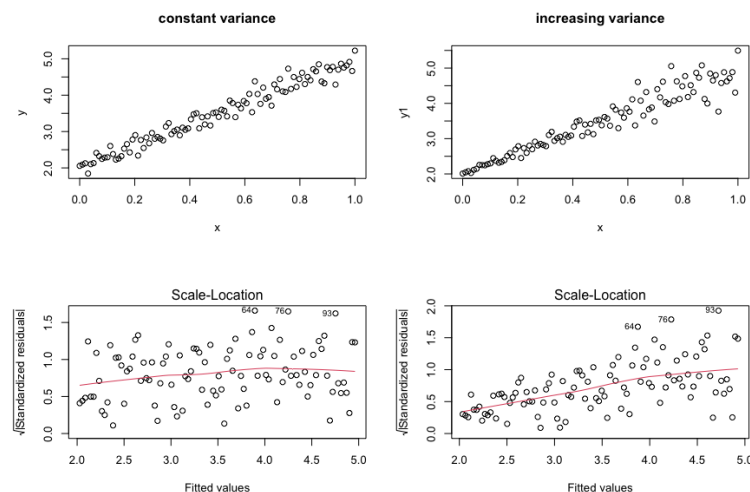


Figure 1.12: The scale-location plot is used to identify failures of the constant-variance assumption. In the right-hand plot, there is a clear upwards trend, showing that larger fitted values are associated with larger residuals.

Residuals vs leverage plot

By plotting standardized residuals against their leverage, as in the right plot of Figure 1.13, it is possible to identify individual observations that may be distorting the fit.

Remark 1.63. We have seen that Cook's distance can be expressed as a function of the standardized residual and the leverage. Hence we can plot contours of Cook's distance on the residuals-vs-leverage plot. As a rule of thumb, a Cook's distance of 1 or more is typically considered large.

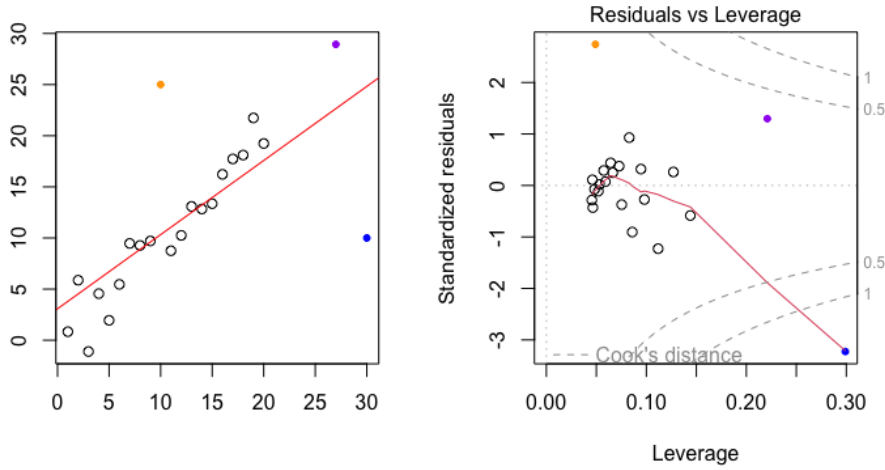


Figure 1.13: The left plot shows a linear regression model. Three atypical observations are highlighted. The residual-leverage plot on the right shows how each observation influences the fit. Dashed contour lines connect points of equal Cook's distance. Only the blue point has distorted the fit, because of the combination of large leverage and a large residual. This is reflected in its substantial Cook's distance.

1.8 The idea of Q-Q plots

In general, if $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, we can write $X_i = \mu + \sigma Z_i$, where $Z_i \sim N(0, 1)$.

If $X_{(i)}$ denotes the i th order statistic, i.e. the i th smallest observation, then we have $X_{(i)} = \mu + \sigma Z_{(i)}$, and so

$$E[X_{(i)}] = \mu + \sigma E[Z_{(i)}].$$

By the probability integral transform, we can write $Z_{(i)} = \Phi^{-1}(U_{(i)})$, where $U_{(i)}$ is the order statistic of a sample from a standard uniform distribution. $U_{(i)}$ has density

$$f_{U_{(i)}}(u) = \frac{n!}{(i-1)!(n-i)!} u^{i-1} (1-u)^{n-i}, \quad u \in (0, 1),$$

so we can calculate

$$E[U_{(i)}] = \int_0^1 \frac{n!}{(i-1)!(n-i)!} u^i (1-u)^{n-i} du = \frac{i}{n+1}.$$

We can understand this result by noticing that n uniform random variables divide the unit interval into $n+1$ segments. By symmetry, these segments have equal expected lengths, which must sum to 1, the total length of the interval, and so must be $\frac{1}{n+1}$. The i th observation is at the end of the i th segment, and so must be $\frac{i}{n+1}$.

A delta method argument can then be used to show that

$$E[Z_{(i)}] \approx \Phi^{-1}\left(E[U_{(i)}]\right) = \Phi^{-1}\left(\frac{i}{n+1}\right).$$

Hence a plot of the ordered data $X_{(i)}$ against $\Phi^{-1}\left(\frac{i}{n+1}\right)$ should give an approximate straight line with intercept μ and gradient σ .

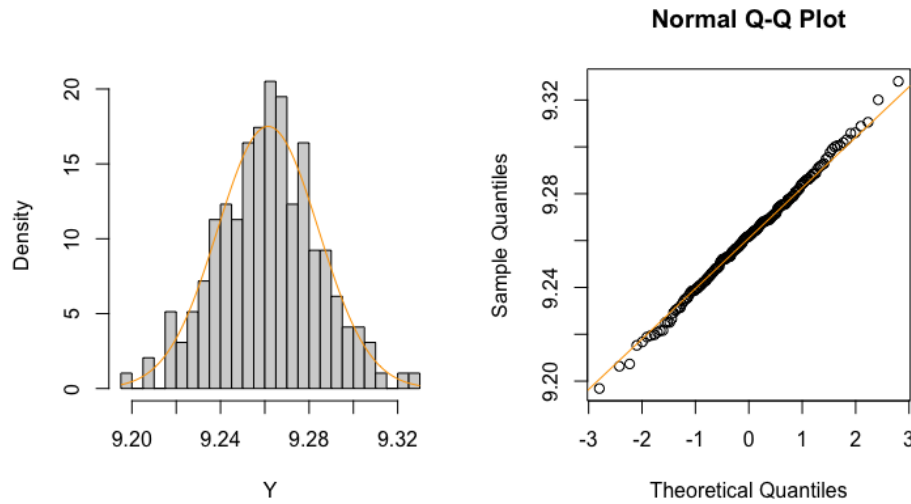


Figure 1.14: Heat flow meter calibration data collected by NIST. Left: orange curve is the best-fitting normal density.

1.9 Smoothing

To criticise the model fit, we often inspect the residuals of the model $\hat{\epsilon}$. We are especially interested in identifying whether there is evidence of a nonlinear relationship between predictor and response that has been missed by the linear model. To do this, we fit a flexible non-parametric model to the residuals. This section gives a brief introduction to a simple method for doing this, called local regression.

We have seen that the fitted values $\hat{y}$ can be written as a linear combination (sometimes called a linear filter) of the observations $y_1, \dots, y_n$.

$$\hat{y}(x_0) = \sum_{i=1}^n w_i(x_0) y_i,$$

where we have suppressed the dependence of the weights on $\mathbf{x} = (x_1, \dots, x_n)$ for simplicity.

Non-parametric approaches for predicting y proceed without assuming an explicit functional form for the systematic relationship between response and predictors.

Example 1.64. One well-known non-parametric method is k -nearest neighbours, where

$$\hat{y}(x_0) = \frac{1}{k} \sum_{i \in N_k(x_0)} y_i,$$

where $N_k(x_0)$ is the set of k nearest observations closest to x_0 in predictor space. The parameter k controls the smoothness of the resulting prediction function: small values of k average over few neighbours, resulting in an irregular prediction, whereas larger values of k yield a smoother prediction.

Remark 1.65. k -nearest neighbours is an intuitive approach, easy to grasp. However, it has undesirable properties: even with a well-chosen value of k , the resulting $\hat{y}$ is often an irregular function, where smoothness would be desirable. This behaviour results from giving equal weight to the k nearest observations, and no weight to all the other observations. A more sophisticated approach is to assign weights to each observation that decay smoothly according to their distance from x_0 .

Definition 1.66. The Nadaraya-Watson kernel-weighted average is

$$\hat{y}(x_0) = \frac{\sum_{i=1}^n K(x_0, x_i) y_i}{\sum_{i=1}^n K(x_0, x_i)},$$

where the kernel function $K = K_\lambda$ depends on a bandwidth parameter λ and a distance function D ,

$$K_\lambda(x_0, x) = D\left(\frac{|x_0 - x|}{\lambda}\right).$$

Remark 1.67. The smoothing parameter λ controls the width of the local neighbourhood. The smoother essentially assumes $\hat{y}$ is constant within the local window. This means that large values of λ impose greater smoothing, at the expense of increasing bias.

Remark 1.68. Various distance functions are in use. A common choice is the tri-cube distance.

Definition 1.69. the tri-cube distance function is

$$D(t) = \begin{cases} (1 - |t|^3)^3 & |t| \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Remark 1.70. The kernel-weighted average can introduce bias, particularly at the boundary of a domain or where x values are not evenly spaced. One way to overcome this is local regression.

Definition 1.71. The locally weighted regression $\hat{y}(x_0) = \hat{\alpha}(x_0) + \hat{\beta}(x_0)x_0$ is the result of minimizing the weighted least squares function

$$\sum_{i=1}^n K_\lambda(x_0, x_i) [y_i - \alpha(x_0) - \beta(x_0)x_i]^2$$

over α and β .

Remark 1.72. The smoother lines produced in R's residual plots are produced using locally weighted regression.

1.10 General linear model

Most commonly, linear models are considered under the assumption of constant variance and independent observations,

$$Y = X\beta + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I).$$

A more general situation arises in practice, which nonetheless reduces to the constant variance case under a change of variables. This is where $\epsilon \sim N(0, \sigma^2 \Sigma)$, where σ^2 remains unknown and Σ is a known positive definite matrix.

Example 1.73. In some contexts, different observations will have systematically different variances, say $Y_i \sim N\left(X_i\beta, \frac{\sigma^2}{w_i}\right)$. For example, if each observation $Y_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Z_j$ is a sample average of n_i independent observations, then $w_i = n_i$ would give the appropriate weighting: observations based on averages of more independent samples will tend to have lower variance. Here Σ is the diagonal matrix with $\Sigma_{ii} = \frac{1}{n_i}$.

Example 1.74. Where data are structured in time, it could be that successive errors are correlated. A simple example might be $\epsilon_{i+1} = \alpha\epsilon_i + \delta_i$, where $\delta_i \sim N(0, \tau^2)$, and where α and τ are constants.

Example 1.75. It could be that observations are grouped in some way, e.g. an agricultural trial comparing yields of different varieties of grain. Suppose the growing area in which the trial takes place is heterogeneous, but can be split into blocks, within each of which the land is comparable. Each variety

sown once within each block and yield recorded. We then use a model where $\text{Cor}(\epsilon_i, \epsilon_j) = \rho$ when i and j are in the same block, and zero otherwise. This is the compound symmetry model.

Proposition 1.76. *The general linear model reduces straightforwardly to the constant variance, independent observation case considered above, as follows. We can always decompose the matrix $\Sigma = SS^T$, where S is upper triangular; this is the **Cholesky decomposition**. Then define $\mathbf{Y}' = S^{-1}\mathbf{Y}$, so that*

$$\begin{aligned}\mathbf{Y}' &= S^{-1}\mathbf{Y} = S^{-1}\mathbf{X}\boldsymbol{\beta} + S^{-1}\boldsymbol{\epsilon} \\ &= \mathbf{X}'\boldsymbol{\beta} + \boldsymbol{\epsilon}',\end{aligned}$$

where

$$\text{Var}(\boldsymbol{\epsilon}') = S^{-1}\text{Var}(\boldsymbol{\epsilon}) \left(S^{-1}\right)^T = S^{-1}\sigma^2 SS^T \left(S^{-1}\right)^T = \sigma^2 I,$$

so that $\boldsymbol{\epsilon} \sim N(0, \sigma^2 I)$, and we can use the standard results from the constant variance linear model to obtain

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= ((\mathbf{X}')^T \mathbf{X}')^{-1} (\mathbf{X}')^T \mathbf{Y}' = \left(\mathbf{X}^T \left(S^{-1}\right)^T S^{-1} \mathbf{X} \right)^{-1} \mathbf{X}^T \left(S^T\right)^{-1} S^{-1} \mathbf{Y} \\ &= \left(\mathbf{X}^T \Sigma^{-1} \mathbf{X} \right)^{-1} \mathbf{X}^T \Sigma^{-1} \mathbf{Y},\end{aligned}$$

with variance

$$\text{Var}(\hat{\boldsymbol{\beta}}) = \left(\mathbf{X}^T \Sigma^{-1} \mathbf{X} \right)^{-1} \sigma^2.$$

Remark 1.77. When we discuss linear mixed models, we will see approaches that be taken in the more challenging but more common situation in which Σ , or its parameters, are also unknown and have to be estimated from the data.

Remark 1.78. Where Σ , or its parameters, are unknown, we use profile likelihood. If the covariance matrix $\Sigma = \Sigma(\boldsymbol{\lambda})$, where $\boldsymbol{\lambda}$ is a vector of parameters, then for fixed $\boldsymbol{\lambda}$, we can compute the estimates $\hat{\boldsymbol{\beta}}(\boldsymbol{\lambda})$ and $\hat{\sigma}^2(\boldsymbol{\lambda})$. We can then use a numerical method to optimise the profile log likelihood

$$l_p(\boldsymbol{\lambda}) = -\frac{n}{2} \log \hat{\sigma}^2(\boldsymbol{\lambda}) - \frac{1}{\hat{\sigma}^2(\boldsymbol{\lambda})} \left(\mathbf{y}' - \mathbf{X}' \hat{\boldsymbol{\beta}}(\boldsymbol{\lambda}) \right)^T \left(\mathbf{y}' - \mathbf{X}' \hat{\boldsymbol{\beta}}(\boldsymbol{\lambda}) \right)$$

over $\boldsymbol{\lambda}$ using a numerical method, e.g. Newton's method.

Remark 1.79. Where the (standardized) residuals of a model have non-constant variance, or are correlated, there are several possible causes. It could be that the dispersion around the line exhibits genuine correlation

or non-constant variance. This is when the general linear model approach makes sense. Alternatively, it could be that the model is misspecified, e.g. an important predictor might be missing, or it could be that the functional form of a predictor in the model is incorrect.

1.11 The Bootstrap

We have seen that inference for the linear model makes assumptions about the probability distribution of the errors ϵ . The bootstrap is a more modern approach. It makes no assumptions about the probability distributions of the errors, but is more computationally expensive. It is popular because for realistic problems, the computational burden is manageable.

The idea of the bootstrap

Suppose we are trying to obtain a confidence interval for a statistic, e.g. the median of a quantity of interest in a population, based only on a random sample, $X_1, \dots, X_n$ from the distribution with CDF F_X . Let M_n be the median of this sample. We proceed by making a large number B of bootstrap samples of size n , by sampling with replacement from the original dataset:

$$\begin{array}{ccc} X_1^{*(1)}, & \dots, & X_n^{*(1)} \\ \vdots & & \vdots \\ X_1^{*(B)}, & \dots, & X_n^{*(B)} \end{array}$$

We then compute the medians $M_n^{*(1)}, \dots, M_n^{*(B)}$ of each bootstrap sample, and use these as a proxy for the true sampling distribution of the median. For example, we can use the sample variance

$$\widehat{\text{Var}}(M_n^*) = \frac{1}{B-1} \sum_{j=1}^B \left(M_n^{*(j)} - \bar{M}_n^* \right)^2; \quad \bar{M}_n^* = \frac{1}{B} \sum_{j=1}^B M_n^{*(j)}$$

as an estimate for the sampling variance of M_n .

Bootstrap confidence intervals

If we were willing to assume M_n is normally distributed, we could then take its standard error to be $\sqrt{\widehat{\text{Var}}(M_n^*)}$ and form a $1 - \alpha$ confidence interval as

$$M_n \pm z_{\frac{\alpha}{2}} \sqrt{\widehat{\text{Var}}(M_n^*)}.$$

If we were unwilling to make this assumption, we could instead use the percentile method: use the $\alpha/2$ and $1 - \alpha/2$ percentile points of the bootstrap distribution as the end points of a confidence interval. It is sometimes

appropriate to make small adjustments to this approach to remove bias , see *Computer Age Statistical Inference*, Chapter 11.

Why bootstrapping works

Let $\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$ be the empirical CDF of the observed data. This is just a step function with a jump at each observed data point. When we sample with replacement from the observed data, we are generating a random variable Z , from the distribution with CDF $\hat{F}_n$, i.e.

$$\Pr(Z = X_i) = \frac{1}{n}, \quad i = 1, 2, \dots, n.$$

Bootstrapping works because $\hat{F}_n \rightarrow F_X$, the true CDF of X , as $n \rightarrow \infty$. Hence we also get convergence for reasonable statistics derived from the distribution of X .

Applied Statistical Inference

Section 2

Generalized Linear Models

2.1 Motivation

Mean-variance relationships

One important property of the linear model we have used so far is that the error variance is constant, and in particular is independent of the mean. This is by no means universally a reasonable assumption: there are many practical settings where the variability in a measurement is a function of what is being measured. A practical example can be seen in Figure 2.1, which displays the mean and variance of the daily number of accidents on Transport for London in each London borough, for each month of 2019.

Three familiar settings occur in statistical work:

- Counts of rare events modelled by the Poisson distribution. If $Y \sim \text{Po}(\mu)$, then

$$\text{E}[Y] = \mu = \text{Var}[Y].$$

- Binomial proportions: If $Y \sim \text{Bin}(n, p)$, then $\text{E}[Y] = np = \mu$, and

$$\text{Var}[Y] = np(1 - p) = \mu \left(1 - \frac{\mu}{n}\right).$$

- Exponential waiting times. If $Y \sim \text{Exp}\left(\frac{1}{\mu}\right)$ then

$$\text{Var}[Y] = \mu^2.$$

These mean-variance relationships must be accounted for if our inference is to be well-calibrated, i.e. if our confidence intervals are to have the coverage we claim, and if our hypothesis tests are to have the claimed significance level and power.

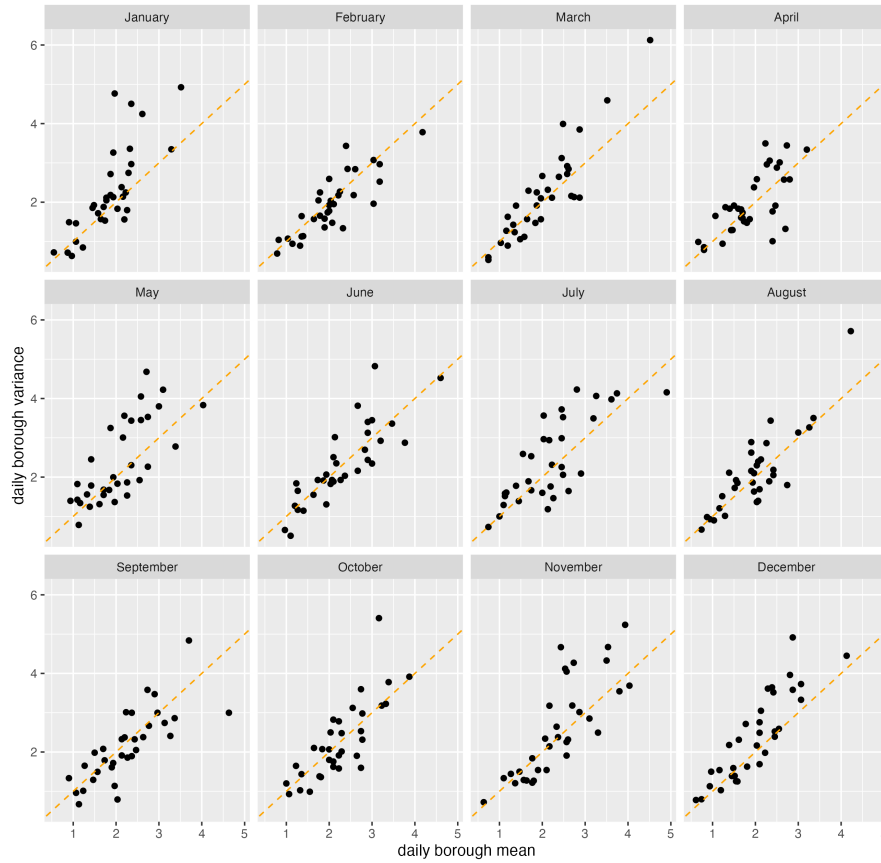


Figure 2.1: Each point in the figure represents a London borough. The plot shows the daily mean number of reported accidents on TfL in the borough against the daily variance in this number. For data such as these, counts of rare events, the Poisson distribution is a good model. One characteristic property is the linear relationship between mean and variance. Data from 2019.

Functional relationships

So far, we have assumed a linear relationship between the response and covariates. But this is somewhat restrictive, and can even be unnatural for distributions whose support is restricted to a segment of the real line. In Figure 2.2, we see a commonly encountered example, Poisson regression, in which the distribution of the response $Y|x \sim \text{Po}(\mu)$, where μ is related to x as

$$\mu = \exp(\beta_0 + \beta_1 x).$$

The linear predictor is related to the mean of the response by a smooth monotonic function, which respects the constraint that $\mu > 0$ for a Poisson variable.

In Figure 2.3, we see a model for data from a designed experiment in which, for each of eight different doses of a toxin, it was required to estimate the proportion of beetles exposed to the toxin that would be killed. We see a pronounced non-linear response between the mean response and dose, as would be expected given that the proportion is bounded between 0 and 1. We

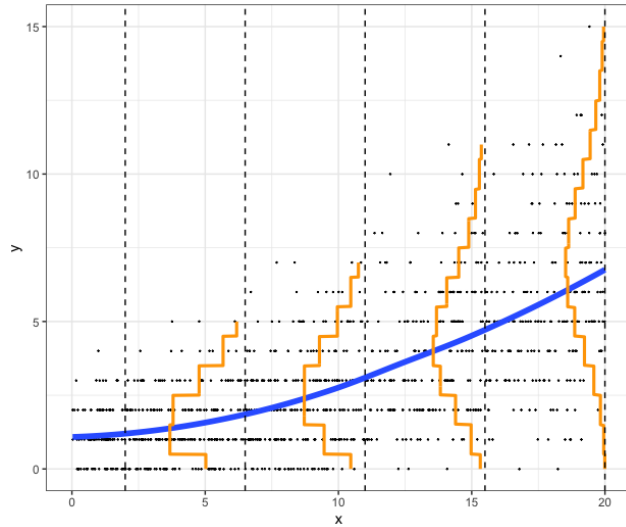


Figure 2.2: In the Poisson regression model, the conditional distribution of $Y|x$ is Poisson with mean $\exp(\beta_0 + \beta_1 x)$, shown in blue. Note that variability around the mean is larger when the mean is larger; a fundamentally different situation from the linear model.

also see a mean-variance relationship: for higher doses, the proportion killed is both higher and less variable. The fitted model accounts for this.

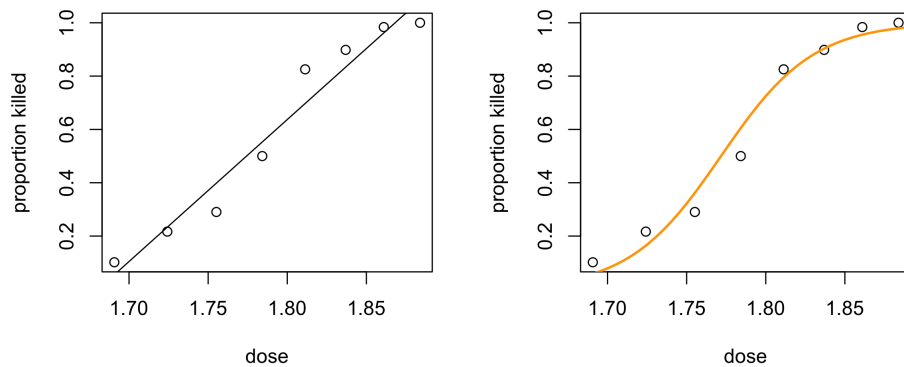


Figure 2.3: Comparison of a linear model and a binomial GLM for the proportion of beetles killed when exposed to a given dose of toxin.

2.2 Specification

NORMAL LINEAR MODELS GENERALIZED LINEAR MODELS

1. **Random Component**

The components of $\mathbf{Y}$ have independent normal distributions with $E(\mathbf{Y}) = \boldsymbol{\mu}$ and common variance σ^2 .

2. **Systematic Component**

Using the covariates $\mathbf{x}_1, \dots, \mathbf{x}_p$ form the linear predictor

$$\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} = \sum_{j=1}^p \mathbf{x}_j \beta_j$$

3. **Link**

The link between the random and systematic components is

$$\boldsymbol{\eta} = \boldsymbol{\mu}$$

1. **Random Component**

The components of $\mathbf{Y}$ are independent and from the same exponential family of distributions with $E(\mathbf{Y}) = \boldsymbol{\mu}$. An observation's variance depends on its mean.

2. **Systematic Component**

Using the covariates $\mathbf{x}_1, \dots, \mathbf{x}_p$ form the linear predictor

$$\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} = \sum_{j=1}^p \mathbf{x}_j \beta_j$$

3. **Link**

The link between the random and systematic components is

$$\eta_i = g(\mu_i) \quad \text{for } i = 1, \dots, n.$$

Remark 2.1. In the specification of a GLM,

- The **random component** defines the probability distribution of the response variable. Specifically, the components of $\mathbf{y}$ have pdf or pmf from an exponential family of distributions, with $E[\mathbf{Y}] = \boldsymbol{\mu}$.
- The **systematic component** specifies a linear predictor $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}$ as a function of the covariates and the unknown parameters.
- The **link function** g can be any monotonic differentiable function. The link function defines the functional relationship between the systematic component and the expectation of the response in the random component; namely $\boldsymbol{\eta} = g(\boldsymbol{\mu})$. For normal linear models the link is the identity function i.e. $g(x) = x$.

Exponential Families

Definition 2.2. Consider a random variable Y whose probability density/mass function depends on the two parameters θ and ϕ . The distribution of Y is said to be a member of an exponential family if its probability density function can be written in the form

$$f_Y(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}, \quad (2.1)$$

for specific functions $b(\cdot)$ and $c(\cdot)$. If ϕ is known, this is an exponential family model with canonical parameter θ .

Remark 2.3. Note that if ϕ is unknown, it is called a nuisance parameter as it will complicate inference of μ or θ , which are usually the parameters of interest. We need to be careful about uncertainty that comes from estimating ϕ when performing hypothesis tests or constructing confidence intervals for the parameters of interest.

Properties of exponential family distributions

Theorem 2.4. The mean and variance of a random variable from an exponential family satisfy

$$\begin{aligned} E[Y] &\equiv \mu = b'(\theta) \\ \text{Var}[Y] &= b''(\theta)\phi \equiv V(\mu)\phi \end{aligned}$$

Proof.

From the definition of a probability density function

$$\int f(y; \theta, \phi) dy = 1.$$

Assuming sufficient regularity that we can pass the derivative through the integral, differentiation with respect to θ gives

$$\frac{\partial}{\partial \theta} \int f(y; \theta, \phi) dy = \int \frac{\partial}{\partial \theta} f(y; \theta, \phi) dy = \frac{\partial}{\partial \theta} 1 = 0. \quad (2.2)$$

If we differentiate again with respect to θ , we get

$$\int \frac{\partial^2}{\partial \theta^2} f(y; \theta, \phi) dy = 0. \quad (2.3)$$

These general results can be applied to distributions from an exponential family. From (2.1) recall that general form of the pdf from an exponential family is:

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\},$$

so that

$$\frac{\partial}{\partial \theta} f(y; \theta, \phi) = \frac{y - b'(\theta)}{\phi} f(y; \theta, \phi).$$

Then by (2.2) we find

$$\int \left(\frac{y - b'(\theta)}{\phi} \right) f(y; \theta, \phi) dy = 0,$$

which simplifies to

$$\frac{1}{\phi} (E[Y] - b'(\theta)) = 0,$$

by definition of the expected value. Thus

$$\mu \equiv E[Y] = b'(\theta). \quad (2.4)$$

Next, we have

$$\frac{\partial^2}{\partial \theta^2} f(y; \theta, \phi) = -\frac{b''(\theta)}{\phi} f(y; \theta, \phi) + \left(\frac{y - b'(\theta)}{\phi} \right)^2 f(y; \theta, \phi). \quad (2.5)$$

Plugging this into (2.3) gives

$$\begin{aligned} 0 &= \int \frac{\partial^2}{\partial \theta^2} f(y; \theta, \phi) dy = -\frac{b''(\theta)}{\phi} + \int \left(\frac{y - b'(\theta)}{\phi} \right)^2 f(y; \theta, \phi) dy \\ &= -\frac{b''(\theta)}{\phi} + \frac{\text{Var}(Y)}{\phi^2} \quad \text{by (2.4)}. \end{aligned}$$

Rearranging yields

$$\text{Var}(Y) = b''(\theta)\phi.$$

Remark 2.5. The function $b''(\theta)$ is referred to as the variance function. The variance function considered as a function of μ will be denoted as $V(\mu)$.

Link Functions

Recall that in the specification of GLMs, the function g describes the link between the expected response $\mu \equiv E(Y)$ and the linear predictor $\eta = X\beta$. We introduced the link function g as

$$\eta_i = g(\mu_i) \quad \text{for } i = 1, \dots, n$$

as any monotonic differentiable function.

Remark 2.6. The link function ensures that the relationship between the linear predictor and the mean is of the correct functional form. Choice of link function is usually a practical modelling question, although one important consideration is the fact that the linear predictor can take any real value, but for some distributions, the mean is restricted to lie in a segment of the real line, e.g. a Poisson mean is strictly positive, and a binomial probability lies between 0 and 1. It is the link function that enforces these constraints.

Definition 2.7. For an exponential family distribution with canonical parameter θ , the canonical link function sets the linear predictor directly equal to the canonical parameter, $\theta = \eta$.

Example 2.8. If $Y \sim \text{Po}(\mu)$, the canonical parameter is $\theta = \log \mu$. The canonical link function sets $\eta = \theta = \log \mu$.

Example 2.9. If $Y \sim \text{Bernoulli}(p)$, the canonical parameter is $\theta = \log \left(\frac{p}{1-p} \right)$.

Remark 2.10. The canonical link is most commonly chosen unless the data generating process naturally suggests an alternative.

2.3 Estimation

Maximum Likelihood Theory

Suppose that $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ has a pdf or pmf $f(\mathbf{y}; \boldsymbol{\psi})$ for $\mathbf{y} \in \mathbf{R}^n$ and parameter vector $\boldsymbol{\psi} \in \mathbf{R}^p$. Then likelihood function is a function of $\boldsymbol{\psi}$ for each fixed $\mathbf{y}$ given by

$$L(\boldsymbol{\psi}; \mathbf{y}) = f(\mathbf{y}; \boldsymbol{\psi}).$$

Similarly, the log-likelihood function is given by

$$\ell(\boldsymbol{\psi}; \mathbf{y}) = \log L(\boldsymbol{\psi}; \mathbf{y}) = \log f(\mathbf{y}; \boldsymbol{\psi}).$$

We then have the following:

$$\begin{aligned} \text{Score Vector: } \mathbf{U}(\boldsymbol{\psi}; \mathbf{y}) &= \nabla \ell(\boldsymbol{\psi}; \mathbf{y}) \quad \text{where } \nabla = \left(\frac{\partial}{\partial \psi_1}, \dots, \frac{\partial}{\partial \psi_p} \right) \\ \text{so that } U_j(\boldsymbol{\psi}; \mathbf{y}) &= \frac{\partial}{\partial \psi_j} \ell(\boldsymbol{\psi}; \mathbf{y}), \quad j = 1, \dots, p \end{aligned}$$

$$\text{Observed Information Matrix: } \mathcal{I} = -\nabla \nabla^T \ell(\boldsymbol{\psi}; \mathbf{y}) \in \mathbf{R}^{p \times p}$$

$$\text{so that } \mathcal{I}_{jk}(\boldsymbol{\psi}; \mathbf{y}) = -\frac{\partial^2}{\partial \psi_j \partial \psi_k} \ell(\boldsymbol{\psi}; \mathbf{y})$$

$$\text{Fisher Information Matrix: } \mathcal{J}(\boldsymbol{\psi}) = \mathbf{E} [\mathcal{I}(\boldsymbol{\psi}; \mathbf{Y})] \in \mathbf{R}^{p \times p}$$

If $Y_1, \dots, Y_n$ are iid with pdf or pmf $f(y_i; \boldsymbol{\psi})$, $y_i \in \mathbf{R}$ then

$$L(\boldsymbol{\psi}; y_1, \dots, y_n) = \prod_{i=1}^n L(\boldsymbol{\psi}, y_i) \quad \text{and} \quad \ell(\boldsymbol{\psi}; y_1, \dots, y_n) = \sum_{i=1}^n \ell(\boldsymbol{\psi}; y_i)$$

Example 2.11. Suppose $Y_1, \dots, Y_n \stackrel{iid}{\sim} \text{Exponential}(1/\mu)$ for some $\mu > 0$. We have

$$\begin{aligned} f(y_i; \mu) &= \frac{1}{\mu} \exp\left(-\frac{y_i}{\mu}\right) \quad y_i \geq 0 \\ \ell(\mu; y_1, \dots, y_n) &= -n \log(\mu) - \sum_{i=1}^n \frac{y_i}{\mu} \\ U = \frac{\partial}{\partial \mu} \ell(\mu; \mathbf{y}) &= -\frac{n}{\mu} + \frac{1}{\mu^2} \sum_{i=1}^n y_i \\ \mathcal{I} = -\frac{\partial^2}{\partial \mu^2} \ell(\mu; y_1, \dots, y_n) &= -\frac{n}{\mu^2} + \frac{2}{\mu^3} \sum_{i=1}^n y_i \\ \mathcal{J} &= -\frac{n}{\mu^2} + \frac{2}{\mu^3} n \mathbf{E}[Y_1] = -\frac{n}{\mu^2} + \frac{2n}{\mu^2} = \frac{n}{\mu^2} \end{aligned}$$

Example 2.12. Suppose $Y_1, \dots, Y_n \sim N(\mu_i, \sigma^2)$ independently with $\sigma^2 > 0$ known and some for $\mu_i \in \mathbf{R}$. We have

$$\begin{aligned} f(y_i; \mu_i) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right) \\ f(y_1, \dots, y_n; \boldsymbol{\mu}) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right) \\ \ell(\boldsymbol{\mu}; y_1, \dots, y_n) &= -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu_i)^2 \\ U_j &= \frac{\partial}{\partial \mu_j} \ell(\boldsymbol{\mu}; \mathbf{y}) = \frac{1}{\sigma^2} (y_j - \mu_j) \\ \mathcal{I}_{jk} &= -\frac{\partial^2}{\partial \mu_j \partial \mu_k} \ell(\boldsymbol{\mu}; \mathbf{y}) = \begin{cases} 1/\sigma^2 & \text{if } j = k \\ 0 & \text{o/w} \end{cases} \\ \mathcal{J}_{jk} &= E[\mathcal{I}_{jk}] = \mathcal{I}_{jk} \end{aligned}$$

Theorem 2.13. Let $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ have a pdf or pmf $f(\mathbf{y}; \boldsymbol{\psi})$ for $\mathbf{y} \in \mathbf{R}^n$ and parameter vector $\boldsymbol{\psi} \in \mathbf{R}^p$. Then, under certain regularity conditions,

$$E[\mathbf{U}] = \mathbf{0} \quad \text{and} \quad \text{Cov}[\mathbf{U}] = E[\mathbf{U}\mathbf{U}^T] = \mathcal{J}.$$

Proof.

For $j = 1, \dots, p$ we have

$$\begin{aligned} E[U_j] &= \int \left(\frac{\partial}{\partial \psi_j} \log f(\mathbf{y}; \boldsymbol{\psi}) \right) f(\mathbf{y}; \boldsymbol{\psi}) d\mathbf{y} \\ &= \int \frac{1}{f(\mathbf{y}; \boldsymbol{\psi})} \left(\frac{\partial}{\partial \psi_j} f(\mathbf{y}; \boldsymbol{\psi}) \right) f(\mathbf{y}; \boldsymbol{\psi}) d\mathbf{y} \\ &= \int \frac{\partial}{\partial \psi_j} f(\mathbf{y}; \boldsymbol{\psi}) d\mathbf{y} \\ &= \frac{\partial}{\partial \psi_j} \int f(\mathbf{y}; \boldsymbol{\psi}) d\mathbf{y} \\ &= \frac{\partial}{\partial \psi_j} 1 = 0 \end{aligned}$$

It then follows that $\text{Cov}[\mathbf{U}] = E[\mathbf{U}\mathbf{U}^T]$ as $E[\mathbf{U}] = \mathbf{0}$. Thus we only need to show that $E[\mathbf{U}\mathbf{U}^T] = \mathcal{J} = E[\mathcal{I}]$. See problem sheet for remainder of proof.

Theorem 2.14. Suppose we have $Y_1, \dots, Y_n$ independent random variables from an exponential family with means $\mu_1, \dots, \mu_n$ and variance functions $V(\mu_1), \dots, V(\mu_n)$ with common dispersion parameter ϕ . Then for $j = 1, \dots, p$ and $k = 1, \dots, p$,

$$U_j = \sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{\text{Var}[Y_i]} x_{ij} \frac{\partial \mu_i}{\partial \eta_i} \right] \quad \text{and} \quad \mathcal{J}_{jk} = \sum_{i=1}^n \left[\frac{x_{ij} x_{ik}}{\text{Var}[Y_i]} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \right].$$

Proof. The likelihood function for $Y_1, \dots, Y_n$ is

$$L(\boldsymbol{\beta}; y_1, \dots, y_n) = \prod_{i=1}^n \exp \left\{ \frac{y_i \theta(\mu_i) - b(\theta(\mu_i))}{\phi} + c(y_i, \phi) \right\}$$

where $\theta(\mu_i) = \theta(g^{-1}(\eta_i))$, $\eta_i = \sum_{j=1}^p x_{ij} \beta_j$. Therefore, the loglikelihood is

$$\ell(\boldsymbol{\beta}; y) = \sum_{i=1}^n \underbrace{\left\{ \frac{y_i \theta(\mu_i) - b(\theta(\mu_i))}{\phi} + c(y_i, \phi) \right\}}_{=: \ell_i} = \sum_{i=1}^n \ell_i$$

For the score function

$$U_j = \frac{\partial \ell(\boldsymbol{\beta}; y)}{\partial \beta_j} = \sum_{i=1}^n \frac{\partial \ell_i}{\partial \beta_j} \quad \text{for } j = 1, \dots, p,$$

we decompose the partial derivative as

$$\frac{\partial \ell_i}{\partial \beta_j} = \frac{\partial \ell_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \beta_j} \quad \text{where } \theta_i \equiv \theta(\mu_i).$$

Now consider each partial derivative in turn. First,

$$\frac{\partial \ell_i}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{\phi} = \frac{y_i - \mu_i}{\phi}$$

Second,

$$\frac{\partial \theta_i}{\partial \mu_i} = \left(\frac{\partial \mu_i}{\partial \theta_i} \right)^{-1} = \frac{1}{b''(\theta_i)} = \frac{\phi}{\text{Var}[Y_i]}.$$

Third,

$$\frac{\partial \mu_i}{\partial \beta_j} = \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j} = \frac{\partial \mu_i}{\partial \eta_i} x_{ij}.$$

Inserting these into the score U_j yields

$$U_j = \sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{\text{Var}[Y_i]} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right], \quad (2.6)$$

as required.

For the Fisher information matrix, $\mathcal{J} = E[\mathbf{U}\mathbf{U}^T]$, we use (2.6) to get

$$\begin{aligned} \mathcal{J}_{jk} &= E \left\{ \sum_{i=1}^n \left[\frac{y_i - \mu_i}{\text{Var}[Y_i]} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right] \sum_{l=1}^n \left[\frac{y_l - \mu_l}{\text{Var}(Y_l)} x_{lk} \left(\frac{\partial \mu_l}{\partial \eta_l} \right) \right] \right\} \\ &= \sum_{i=1}^n \left[\frac{E\{(Y_i - \mu_i)^2\}}{\text{Var}[Y_i]^2} x_{ij} x_{ik} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \right] \\ &= \sum_{i=1}^n \left[\frac{x_{ij} x_{ik}}{\text{Var}[Y_i]} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \right]. \end{aligned}$$

where we used that the observations are independent. $\square$

These results can be succinctly be written as

$$\mathbf{U} = \mathbf{X}^T \mathbf{W} \mathbf{c}, \quad (2.7)$$

where the components of $\mathbf{c}$ are given by

$$c_i = (y_i - \mu_i) \frac{\partial \eta_i}{\partial \mu_i},$$

and the Fisher information matrix can be represented as

$$\mathcal{J} = \mathbf{X}^T \mathbf{W} \mathbf{X}, \quad (2.8)$$

with $\mathbf{W} \in \mathbf{R}^{n \times n}$ being a diagonal matrix with entries

$$w_{ii} = \frac{1}{\text{Var}[Y_i]} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2.$$

Remark 2.15. To find the MLE, $\hat{\boldsymbol{\beta}}$, we solve $\nabla \ell = 0$. This is equivalent to solving $U_j = 0$ for $j = 1, \dots, p$. Therefore, the maximum likelihood equations are

$$\sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{\text{Var}[Y_i]} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right] = 0, \quad \text{for } j = 1, \dots, p. \quad (2.9)$$

Note that since $\text{Var}[Y_i] = \phi V(\mu)$, we can write these equations as

$$\sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{V(\mu_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right] = 0, \quad \text{for } j = 1, \dots, p. \quad (2.10)$$

In this form, the equations do not depend on ϕ - this is parallel to maximum likelihood estimation of $\boldsymbol{\beta}$ in the normal linear model, which can proceed without making an estimate of σ^2 .

Remark 2.16. In general, this is a non-linear system of equations in $\boldsymbol{\beta}$ and a numerical solution is required. We proceed to solve (2.10) using a variant of Newton's method.

Algorithm. Newton's method: For $f(\mathbf{x}) : \mathcal{X} \subseteq \mathbf{R}^p \rightarrow \mathbf{R}; \mathbf{x} \mapsto f(\mathbf{x})$, the algorithm below:

$$\mathbf{x}^{(m+1)} = \mathbf{x}^{(m)} - H^{-1} \mathbf{g}, \quad \text{where } H = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} \Big|_{\mathbf{x}=\mathbf{x}^{(m)}}, \quad \mathbf{g} = \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} \Big|_{\mathbf{x}=\mathbf{x}^{(m)}}$$

will converge to $\mathbf{x}^* = \underset{\mathbf{x}}{\text{argmax}} f(\mathbf{x})$ (under regularity conditions).

Using Newton's method for $\ell(\boldsymbol{\beta}; \mathbf{y})$ yields:

$$\hat{\boldsymbol{\beta}}^{(m+1)} = \hat{\boldsymbol{\beta}}^{(m)} + \left(\mathcal{I}^{(m)} \right)^{-1} \mathbf{U}^{(m)},$$

where $\mathbf{U}^{(m)}$ is the score vector evaluated at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}^{(m)}$, and similarly for the observed information matrix $\mathcal{I}^{(m)}$. In maximum likelihood estimation it is

common to estimate $\mathcal{I}$ by $E(\mathcal{I}) = \mathcal{J}$ i.e. the Fisher information matrix¹.
The result is the Fisher scoring algorithm:

¹ see problem sheet

$$\hat{\beta}^{(m+1)} = \hat{\beta}^{(m)} + \left(\mathcal{J}^{(m)} \right)^{-1} \mathbf{u}^{(m)}. \quad (2.11)$$

Continuing to find simpler representations for the iterations of the scoring algorithm:

Plugging in (2.7) and (2.8) into (2.11) gives

$$\begin{aligned} \hat{\beta}^{(m+1)} &= \hat{\beta}^{(m)} + \left(X^T W^{(m)} X \right)^{-1} X^T W^{(m)} \mathbf{c}^{(m)} \\ &= \underbrace{\left(X^T W^{(m)} X \right)^{-1} \left(X^T W^{(m)} X \right)}_{\text{identity}} \hat{\beta}^{(m)} + \left(X^T W X^{(m)} \right)^{-1} X^T W^{(m)} \mathbf{c}^{(m)} \\ &= \left(X^T W^{(m)} X \right)^{-1} X^T W^{(m)} \underbrace{\left\{ X \hat{\beta}^{(m)} + \mathbf{c}^{(m)} \right\}}_{:= \mathbf{z}^{(m)}} \\ \text{or } \left(X^T W^{(m)} X \right) \hat{\beta}^{(m+1)} &= X^T W^{(m)} \mathbf{z}^{(m)}. \end{aligned} \quad (2.12)$$

These iterated weighted least squares (IWLS) equations (2.12) may be viewed as estimating the parameters of a sequence of general linear models as in 1.10.

$$\mathbf{z} = X\beta + \mathbf{c}, \quad \text{where } \mathbf{c} = (\mathbf{y} - \mu)(\partial\eta/\partial\mu) \sim N(\mathbf{0}, W^{-1}). \quad (2.13)$$

Remark 2.17. As we have seen in Section 1.10, we can reduce this to estimation of a linear model by rewriting (2.13) as

$$\mathbf{z} = X\beta + W^{-1/2}\epsilon.$$

Then multiplying on the left by $W^{1/2}$

$$\underbrace{W^{1/2}\mathbf{z}}_{\tilde{\mathbf{y}}} = \underbrace{W^{1/2}X}_{\tilde{X}}\beta + \epsilon, \quad \epsilon \sim N(0, I_n),$$

which is of the form of a normal linear model. We can write the estimate down directly

$$\begin{aligned} \hat{\beta} &= (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \tilde{\mathbf{y}} \\ &= (X^T W X)^{-1} X^T W \mathbf{z}, \end{aligned}$$

which is similar to (2.12). This motivates the iterative weighted least squares (IWLS) algorithm.

Remark 2.18. The method works by approximating the log likelihood by its second order Taylor expansion about a known estimate of the maximum. This is a quadratic form in the parameters. The maximum of this quadratic form is easy to find, and is taken as an improved estimate of the maximum, which then fed back into the method.

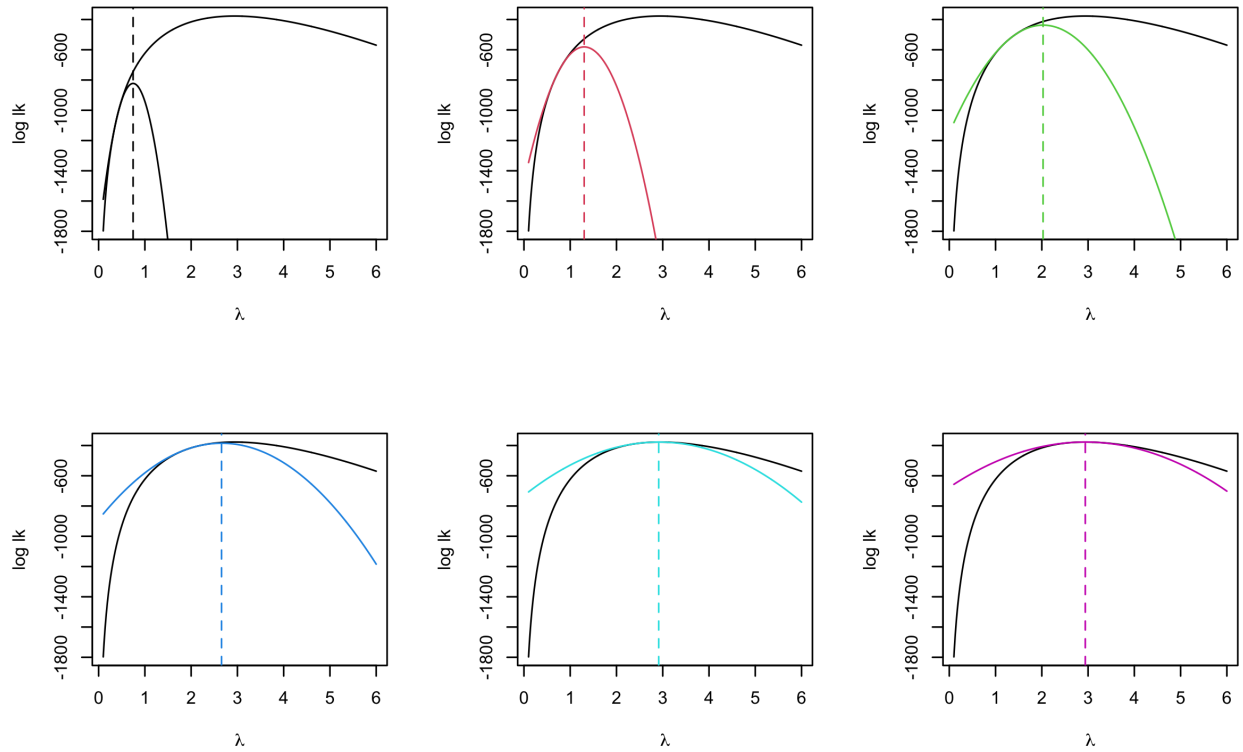


Figure 2.4: Six iterations of Newton's method to find the maximum likelihood estimate of a parameter λ . At each state, the method approximates the log likelihood by the best fitting parabola, finds its vertex, and uses this point as a better approximation to the location of the maximum.

Algorithm 2.1 Iterative weighted least squares (IWLS) algorithm

- 1: Given a current estimate $\hat{\beta}$, form the linear predictor $\hat{\eta}$ and the fitted values $\hat{\mu}$.
- 2: Form the adjusted dependent variable

$$z_i = \hat{\eta}_i + (y_i - \hat{\mu}_i) \frac{\partial \eta}{\partial \mu} \bigg|_{\mu = \hat{\mu}_i}$$

- 3: Form the adjusted weights $\tilde{w}_{ii}$ by

$$\tilde{w}_{ii}^{-1} = \left(\frac{\partial \eta}{\partial \mu} \right)^2 V(\mu) \bigg|_{\mu = \hat{\mu}_i}$$

- 4: Regress z_i on x_i with weights $\tilde{w}_{ii}$ and obtain the new estimate $\hat{\beta}$.
 - 5: Repeat steps 1 to 4 until convergence.
-

Remark 2.19. The adjusted weights are given by $\tilde{w}_{ii} = \phi w_{ii}$. These allow us to do IWLS without estimating ϕ , as noted in 2.15.

Remark 2.20. In addition to providing an estimate of the model parameters, IWLS also gives an estimate of the variance-covariance matrix, as follows.

$$\begin{aligned}
 \text{Cov}[\hat{\beta}] &= \text{Cov} \left\{ \left(X^T W X \right)^{-1} X^T W z \right\} \\
 &= \left[\left(X^T W X \right)^{-1} X^T W \right] \underbrace{\text{Cov}(z)}_{W^{-1}} \left[\left(X^T W X \right)^{-1} X^T W \right]^T \\
 &= \left(X^T W X \right)^{-1} = \phi (X^T \tilde{W} X)^{-1} \\
 &= \mathcal{J}^{-1}.
 \end{aligned} \tag{2.14}$$

Notice, that $\mathcal{J}$ involves the dispersion parameter ϕ — this needs to be known or estimated in order to compute $\mathcal{J}$; see Section 2.41. We estimate $\text{Cov}(\hat{\beta})$ by “plugging” in the last $\tilde{W}^{(m)}$ from the IWLS algorithm and an estimate $\hat{\phi}$ of ϕ to get

$$\text{Cov}[\hat{\beta}] \approx \hat{\phi} (X^T \tilde{W}^{(m)} X)^{-1}$$

2.4 Inference

Here we will be interested in obtaining confidence intervals for individual model parameters, or testing hypotheses corresponding to nested models. In particular we will only be comparing models from the same exponential family and with the same link function. In typical applications, the exponential family to use for the response, and the link function will be determined by the data context.

Sampling Distributions for GLMs

Remark 2.21. For GLMs, in the large-sample regime, it will often be the case that a statistic of interest S , satisfies

$$\frac{S - E[S]}{\sqrt{\text{Var}[S]}} \rightsquigarrow N(0, 1)$$

or equivalently

$$\frac{(S - E[S])^2}{\text{Var}[S]} \rightsquigarrow \chi_1^2,$$

where we shall use $\rightsquigarrow$ to denote “approximately distributed as”, with the approximation becoming closer in the limit of large n .

If there is a vector of statistics of interest

$$S = \begin{pmatrix} S_1 \\ \vdots \\ S_p \end{pmatrix}$$

with asymptotic expectation $E[S]$ and asymptotic variance-covariance matrix Q , then asymptotically

$$S - E[S] \rightsquigarrow N(\mathbf{0}, Q) \quad (2.15)$$

and further, asymptotically

$$(S - E[S])^T Q^{-1} (S - E[S]) \rightsquigarrow \chi_p^2 \quad (2.16)$$

provided Q is non-singular so that Q^{-1} exists and is unique.

Remark 2.22. Where they hold, asymptotic approximations follow from the central limit theorem. In any practical context, it is important to check, e.g. by simulation, that the asymptotic approximations are acceptably close.

Central Limit Theorem

Theorem 2.23. Suppose $A_1, \dots, A_n$ are iid random variables with $E(A_i) = \mu$ and $\text{Var}(A_i) = \sigma^2$ (both finite). Then

$$\sqrt{n} \left(\frac{\frac{1}{n} \sum_{i=1}^n A_i - \mu}{\sqrt{\sigma^2}} \right) \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty$$

where $\xrightarrow{d}$ means convergence in distribution.

Multivariate Central Limit Theorem

Theorem 2.24. If $A_1, \dots, A_n$ are iid random vectors with $E(A_i) = \mu \in \mathbf{R}^p$ and finite, positive definite, symmetric covariance matrix Q then

$$\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n A_i - \mu \right) \xrightarrow{d} N(\mathbf{0}, Q) \quad \text{as } n \rightarrow \infty$$

The asymptotic chi-squared result (2.16) follows from the fact: If $\mathbf{Z} \sim N(\mu, Q)$ where $\mu \in \mathbf{R}^p$ and $Q \in \mathbf{R}^{p \times p}$ is a positive definite, symmetric matrix, then

$$(\mathbf{Z} - \mu)^T Q^{-1} (\mathbf{Z} - \mu) \sim \chi_p^2.$$

Sampling distribution of the score

We can apply the ideas above to the score vector $\mathbf{U}$. Recall from (2.6) that

$$U_j = \frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{\text{Var}[Y_i]} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right], \quad j = 1, \dots, p.$$

Since $E[\mathbf{U}] = \mathbf{0}$ and $\text{Cov}[\mathbf{U}] = \mathcal{J}$ we have, from (2.15) asymptotically

$$\mathbf{U} \rightsquigarrow N(\mathbf{0}, \mathcal{J}) \quad (2.17)$$

and, from (2.16), asymptotically,

$$\mathbf{U}^T \mathcal{J}^{-1} \mathbf{U} \rightsquigarrow \chi_p^2 \quad (2.18)$$

Example Suppose $Y_1, \dots, Y_n$ are independent and $Y_i \sim N(\mu, \sigma^2)$ where μ is the parameter of interest and $\sigma^2 > 0$ is a known constant. The log-likelihood function is

$$\ell(\boldsymbol{\beta}; \mathbf{y}) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 - n \log(\sigma\sqrt{2\pi}).$$

Then the score is

$$U = \frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \mu) = \frac{n}{\sigma^2} (\bar{Y} - \mu),$$

and the Fisher information matrix is

$$\mathcal{J} = \text{Var}[U] = \frac{1}{\sigma^4} \sum_{i=1}^n \text{Var}[Y_i] = \frac{n}{\sigma^2} \quad \text{as} \quad \text{Var}[Y_i] = \sigma^2.$$

Then by rearranging (2.17), we get

$$\bar{Y} \sim N(\mu, \sigma^2/n).$$

As we have seen in Example 1.35, can use this result to construct a confidence interval for μ . For example, a 95% confidence interval for μ is $\bar{y} \pm 1.96\sigma/\sqrt{n}$. In this case this is exact, rather than an asymptotic approximation. ■

Sampling distribution for MLEs

Let $\hat{\boldsymbol{\beta}}$ be the MLE for $\boldsymbol{\beta}$ in a GLM i.e.

$$\hat{\boldsymbol{\beta}} := \underset{\boldsymbol{\beta}}{\operatorname{argmax}} \ell(\boldsymbol{\beta})$$

and so $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \frac{\partial \ell(\hat{\boldsymbol{\beta}})}{\partial \boldsymbol{\beta}} = \mathbf{0}$.

Now consider the 1st order Taylor expansion of $\mathbf{U}(\boldsymbol{\beta})$ about the MLE $\hat{\boldsymbol{\beta}}$:

$$\mathbf{U}(\boldsymbol{\beta}) \approx \underbrace{\mathbf{U}(\hat{\boldsymbol{\beta}})}_{=\mathbf{0}} + \mathbf{U}'(\hat{\boldsymbol{\beta}})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \approx -\mathcal{J}(\hat{\boldsymbol{\beta}})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}),$$

where in the large sample limit we have approximated $\mathbf{U}'$ by $E[\mathbf{U}'] = -\mathcal{J}$. Consequently, we find

$$(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left\{ \mathcal{J}(\hat{\boldsymbol{\beta}}) \right\}^{-1} \mathbf{U}(\boldsymbol{\beta}),$$

assuming that $\mathcal{J}$ is invertible. If $\mathcal{J}$ is regarded as a constant then $\hat{\beta}$ is unbiased for β . It turns out that this is at least asymptotically true.

The asymptotic variance-covariance matrix for $\hat{\beta}$ is

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T] &= E \left[\mathcal{J}^{-1}(\hat{\beta}) \mathbf{u}(\beta) \mathbf{u}(\beta)^T \underbrace{(\mathcal{J}^{-1}(\hat{\beta}))^T}_{\mathcal{J}^{-1}} \right] \\ &= \mathcal{J}^{-1} E \left[\underbrace{\mathbf{u}(\beta) \mathbf{u}(\beta)^T}_{\mathcal{J}} \right] \mathcal{J}^{-1} \\ &= \mathcal{J}(\hat{\beta})^{-1}, \end{aligned}$$

where we have used that $\mathcal{J} = \mathcal{J}^T$ and treated $\mathcal{J}$ as a constant (in the large sample limit).

Thus, by (2.15), we have

$$\hat{\beta} \rightsquigarrow N(\beta, \mathcal{J}^{-1}). \quad (2.19)$$

and also, using (2.16), we have

$$(\hat{\beta} - \beta)^T \mathcal{J}(\hat{\beta})(\hat{\beta} - \beta) \rightsquigarrow \chi_p^2. \quad (2.20)$$

Definition 2.25. The quadratic form in $(\hat{\beta} - \beta)$ given in 2.20 is called the Wald statistic.

Remark 2.26. We can use (2.19) to construct confidence intervals and conduct hypothesis tests involving the components of β . For instance, from (2.19), we have

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1)}} \rightsquigarrow N(0, 1), \quad \text{where} \quad \text{Var}(\hat{\beta}_1) = \underbrace{(X^T W X)^{-1}_{11}}_{\text{the 1,1 entry of } (X^T W X)^{-1}}$$

which we can use to construct confidence intervals for β_1 . Further, under the null hypothesis $H_0 : \beta_1 = 0$ we have $\frac{\hat{\beta}_1}{\sqrt{\text{Var}(\hat{\beta}_1)}} \rightsquigarrow N(0, 1)$.

2.5 Prediction

For a given column vector of covariates $\mathbf{x}_*$, we have $\hat{\eta}_* = \mathbf{x}_*^T \hat{\beta}$ with variance $\mathbf{x}_*^T (X^T W X)^{-1} \mathbf{x}_*$. We can use this to construct an approximate confidence interval for $\hat{\eta}_*$, since $\hat{\eta}_*$ must be approximately normally distributed, because it is a linear combination of the entries of $\hat{\beta}$.

However, we are usually more interested in having a confidence interval in terms of μ . To obtain this, we apply g^{-1} , the inverse of the link function, to the end points of the confidence interval for η_* . More precisely, an approximate 95% confidence interval for the mean response of a prediction with

covariates $\mathbf{x}_\star$ is

$$\left(g^{-1} \left(\hat{\eta}_\star - 1.96 \sqrt{\mathbf{x}_\star^T (X^T W X)^{-1} \mathbf{x}_\star} \right), g^{-1} \left(\hat{\eta}_\star + 1.96 \sqrt{\mathbf{x}_\star^T (X^T W X)^{-1} \mathbf{x}_\star} \right) \right).$$

2.6 Goodness of Fit

One natural way to evaluate the goodness of fit of a model is to compare its log likelihood to the best possible log likelihood, obtained by fitting a more general model from the same family with as many free parameters as can possibly be fitted.

Definition 2.27. A model with as many free parameters as observations is said to be saturated .

Remark 2.28. We can specify a saturated GLM for a sample of n independent observations in terms of the canonical parameters $\theta_1, \dots, \theta_n$ of the individual observations.

Proposition 2.29. When fitting a GLM with n free parameters to n observations, the log likelihood is maximized for $\hat{\mu}_i = y_i$, for $i = 1, \dots, n$, where $\hat{\mu}_i = b'(\hat{\theta}_i)$ and $\hat{\theta}_i$ is the maximum likelihood estimate of the i th canonical parameter.

Proof. For n observations from a GLM with n parameters, the log-likelihood is

$$\ell(\boldsymbol{\theta}, \boldsymbol{\phi}; \mathbf{y}) = \sum_{i=1}^n \left(\frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right).$$

Then, for $i = 1, \dots, n$,

$$\frac{\partial \ell}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{\phi} = \frac{y_i - \mu_i}{\phi}$$

From this, it is clear that log-likelihood is maximised if we force $\mu_i = y_i$ for $i = 1, \dots, n$.

Now write the log-likelihood in terms of the mean response $\boldsymbol{\mu}$:

$$\ell(\boldsymbol{\mu}, \boldsymbol{\phi}; \mathbf{y}) = \sum_{i=1}^n \left(\frac{y_i \theta(\mu_i) - b(\theta(\mu_i))}{\phi} + c(y_i, \phi) \right).$$

Thus, the maximum achievable log-likelihood is $\ell(\mathbf{y}, \boldsymbol{\phi}; \mathbf{y})$.

Remark 2.30. For any model of interest, which will be a GLM with fewer parameters than observations ($p < n$), we can maximise the log-likelihood to get $\hat{\boldsymbol{\beta}}$, then $\hat{\boldsymbol{\eta}} = X\hat{\boldsymbol{\beta}}$ and so the estimated mean response $\hat{\boldsymbol{\mu}} = g^{-1}(\hat{\boldsymbol{\eta}})$ — leading the maximised log-likelihood $\ell(\hat{\boldsymbol{\mu}}, \boldsymbol{\phi}; \mathbf{y})$.

Deviance

Our usual measure of the goodness of fit of a GLM is the deviance.

Definition 2.31. The deviance for a model with estimated mean response $\hat{\mu}$ is

$$D = 2\phi \{ \ell(\mathbf{y}, \phi; \mathbf{y}) - \ell(\hat{\mu}, \phi; \mathbf{y}) \},$$

and the scaled deviance is $D^* = D/\phi$.

Remark 2.32. Since the difference of log likelihoods depends on ϕ only through a multiplicative factor of $\frac{1}{\phi}$, the deviance has no functional dependence on ϕ and can be computed without estimating ϕ . To compute the scaled deviance, we need to know (or estimate) ϕ .

Remark 2.33. In what follows, we will typically write the log-likelihood in terms of β , so that e.g. we write the scaled deviance as

$$D^* = 2 \left\{ \ell(\hat{\beta}_{\text{sat}}; \mathbf{y}) - \ell(\hat{\beta}; \mathbf{y}) \right\},$$

where $\hat{\beta}_{\text{sat}}$ is the MLE of β_{sat} for the saturated model and $\hat{\beta}$ is still the MLE of β in the model of interest.

Remark 2.34. Strictly, there is not just a single saturated model: we could specify a saturated model by choosing as variables any basis for the n -dimensional column space of X . However, as we have seen, any saturated model will lead to $\hat{\mu}_i = y_i$ for $i = 1, 2, \dots, n$.

Example 2.35. Consider $Y_1, \dots, Y_n$ independent with $Y_i \sim N(\mu_i, \sigma^2)$ and

$$E[Y_i] = \mu_i = \sum_{j=1}^p x_{ij}\beta_j.$$

The log-likelihood function is

$$\ell(\mu, \sigma^2; \mathbf{y}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mu)^T (\mathbf{y} - \mu).$$

Setting $\mu = \mathbf{y}$ yields the maximum achievable log-likelihood:

$$\ell(\mathbf{y}, \sigma^2; \mathbf{y}) = -\frac{n}{2} \log(2\pi\sigma^2).$$

Then for any other model with $p < n$ (not a saturated model) we find the maximum likelihood estimate $\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}$ — which in turn gives the estimated mean response $\hat{\mu} = g^{-1}(\hat{\eta}) = X\hat{\beta}$. So

$$\ell(\hat{\beta}, \sigma^2; \mathbf{y}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - X\hat{\beta})^T (\mathbf{y} - X\hat{\beta}).$$

Therefore, recalling that $\phi = \sigma^2$ for the normal family, the deviance is

$$\begin{aligned} D &= 2\sigma^2 \left\{ \ell(\hat{\beta}_{\text{sat}}, \sigma^2; \mathbf{y}) - \ell(\hat{\beta}, \sigma^2; \mathbf{y}) \right\} \\ &= 2\sigma^2 \left\{ -\frac{n}{2} \log(2\pi\sigma^2) - \left[-\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - X\hat{\beta})^T (\mathbf{y} - X\hat{\beta}) \right] \right\} \\ &= (\mathbf{y} - X\hat{\beta})^T (\mathbf{y} - X\hat{\beta}) \\ &= (\mathbf{y} - \hat{\mathbf{y}})^T (\mathbf{y} - \hat{\mathbf{y}}) \quad \text{which we recognise as the RSS.} \end{aligned}$$

Example 2.36. For a Poisson GLM, we have

$$\ell(\boldsymbol{\mu}; \mathbf{y}) = \sum_{i=1}^n -\mu_i + y_i \log \mu_i - \log y_i!.$$

For this exponential family, the dispersion parameter is $\phi = 1$. Hence the deviance is just given by

$$D = 2 \{ \ell(\mathbf{y}; \mathbf{y}) - \ell(\hat{\boldsymbol{\mu}}; \mathbf{y}) \} = 2 \sum_{i=1}^n y_i \log \left(\frac{y_i}{\hat{\mu}_i} \right) - y_i + \hat{\mu}_i.$$

Definition 2.37. The Pearson X^2 statistic for a model is

$$X^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}.$$

Remark 2.38. The Pearson X^2 is another commonly used measure of the discrepancy of a GLM. Pearson's X^2 statistic shares similar asymptotic distribution properties with the (scaled) deviance. However, in this module we shall focus on the deviance as a measure of goodness-of-fit as

- the deviance has a general advantage as a discrepancy measure in that it is additive for nested sets of models (see next section).
- the deviance leads to better normalised residuals.

Hypothesis Testing

We now discuss how to compare GLMs by hypothesis testing. These tests will involve the deviance and scaled deviance, therefore we first need their sampling distributions.

Proposition 2.39. For large samples, the scaled deviance $D^* \sim \chi^2(n - p)$.

Proof. First, consider the 2nd order Taylor expansion of the log-likelihood around the MLE $\hat{\beta}$:

$$\begin{aligned} \ell(\boldsymbol{\beta}) &\approx \ell(\hat{\boldsymbol{\beta}}) + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \underbrace{\frac{\partial \ell}{\partial \boldsymbol{\beta}} \Big|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}}_{=\mathbf{U}(\hat{\boldsymbol{\beta}})=0} + \frac{1}{2} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \underbrace{\frac{\partial^2 \ell}{\partial \boldsymbol{\beta}^2} \Big|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}}_{=\mathbf{U}'(\hat{\boldsymbol{\beta}}) \approx E(\mathbf{U}') = -\mathcal{J}} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \\ &\approx \ell(\hat{\boldsymbol{\beta}}) - \frac{1}{2} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \mathcal{J} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}), \end{aligned}$$

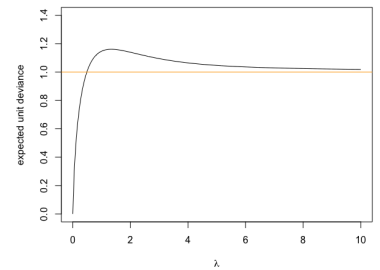


Figure 2.5: For $Y \sim \text{Po}(\lambda)$, the expected unit deviance tends to 1 as $\lambda \rightarrow \infty$, reflecting the convergence of the deviance to a $\chi^2(1)$ variable.

where we approximated $\mathbf{U}'$ by $E[\mathbf{U}'] = -\mathcal{J}$. Consequently, we find

$$2 \left\{ \ell(\hat{\boldsymbol{\beta}}) - \ell(\boldsymbol{\beta}) \right\} = (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \mathcal{J}(\hat{\boldsymbol{\beta}}) (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \quad (2.21)$$

which is approximately χ_p^2 distributed by (2.20). Rewriting the deviance and using (2.21) gives

$$\begin{aligned} D^* &= 2 \left\{ \ell(\hat{\boldsymbol{\beta}}_{\text{sat}}; \mathbf{y}) - \ell(\hat{\boldsymbol{\beta}}; \mathbf{y}) \right\} \\ &= 2 \underbrace{\left\{ \ell(\hat{\boldsymbol{\beta}}_{\text{sat}}; \mathbf{y}) - \ell(\boldsymbol{\beta}_{\text{sat}}; \mathbf{y}) \right\}}_{\approx \chi_n^2} - 2 \underbrace{\left\{ \ell(\hat{\boldsymbol{\beta}}; \mathbf{y}) - \ell(\boldsymbol{\beta}; \mathbf{y}) \right\}}_{\approx \chi_p^2} + \underbrace{2 \left\{ \ell(\boldsymbol{\beta}_{\text{sat}}; \mathbf{y}) - \ell(\boldsymbol{\beta}; \mathbf{y}) \right\}}_{:= v > 0 \text{ positive constant}} \end{aligned}$$

If the model with p parameters is correct, and we assume that v is negligible, then

$$D^* \approx \chi_{n-p}^2.$$

Remark 2.40. It is more accurate to say that for finite samples, D^* has a non-central $\chi^2(n-p, v)$ distribution: v is the non-centrality parameter. For large n , $v \rightarrow 0$, so that the distribution of D^* approaches a $\chi^2(n-p)$.

Model Comparison with known ϕ

Now consider comparing two nested models as follows: Consider the null hypothesis

$$H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0 = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_q \end{pmatrix},$$

corresponding to model M_0 and a more general hypothesis, the alternative,

$$H_1 : \boldsymbol{\beta} = \boldsymbol{\beta}_1 = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix},$$

corresponding to model M_1 with $q < p < n$. We can test H_0 against H_1 using the difference of the scaled deviances

$$\begin{aligned} D_0^* - D_1^* &= 2 \left\{ \ell(\hat{\boldsymbol{\beta}}_{\text{sat}}; \mathbf{y}) - \ell(\hat{\boldsymbol{\beta}}_0; \mathbf{y}) \right\} - 2 \left\{ \ell(\hat{\boldsymbol{\beta}}_{\text{sat}}; \mathbf{y}) - \ell(\hat{\boldsymbol{\beta}}_1; \mathbf{y}) \right\} \\ &= 2 \left\{ \ell(\hat{\boldsymbol{\beta}}_1; \mathbf{y}) - \ell(\hat{\boldsymbol{\beta}}_0; \mathbf{y}) \right\}. \end{aligned}$$

The statistic $D_0^* - D_1^*$ is approximately χ_{p-q}^2 distributed under the null hypothesis.

Model Comparison with unknown ϕ

Under the null hypothesis we have

$$D_1^* \sim \chi_{n-p}^2 \quad \text{and} \quad D_0^* - D_1^* \sim \chi_{p-q}^2.$$

Taking D_1^* and $D_0^* - D_1^*$ as asymptotically independent, then

$$\frac{(D_0^* - D_1^*) / (p - q)}{D_1^* / (n - p)} \sim F_{(p-q), (n-p)}.$$

The advantage of this test statistic is that we can multiply top and bottom by ϕ to get a test statistic based on the deviance:

$$\frac{(D_0 - D_1) / (p - q)}{D_1 / (n - p)} \sim F_{(p-q), (n-p)}.$$

The advantage of this hypothesis test for model comparison is that it does not depend on ϕ .

Remark 2.41. (Warning) There are certain assumptions made when proving that the deviance, D , is approximately or asymptotically χ_{n-p}^2 distributed to bear in mind,

- The observations are independent and distributed according to some member of an exponential family.
- The approximation relies on the number of parameters in the model staying fixed, while the sample size tends to infinity. But the saturated model has as many parameters as number of data. Also, in the Binomial case the χ_{n-p}^2 approximation (or asymptotics) for the deviance are based upon the following assumptions
 - the observations are truly distributed independently according to the binomial distribution;
 - The approximation is based on a limiting operation in which n is fixed and $n_i \rightarrow \infty$ for each i (and in fact $n_i \pi_i (1 - \pi_i) \rightarrow \infty$).

However, the χ^2 approximation is ordinarily reliable when comparing two nested models, as the deviance for the saturated model cancels out. In practice, one can always check by simulation whether the asymptotics are reliable.

Estimating the Dispersion Parameter

Recall that the MLE for β does not depend on the dispersion ϕ . However, in cases where the dispersion is unknown, it must be estimated before inference can be carried out. There are two commonly used estimators:

Definition 2.42. The deviance estimator of the dispersion parameter is

$$\hat{\phi}_D = \frac{D}{n - p},$$

where D is the deviance of the model.

Remark 2.43. This is essentially a moment estimator, based on the asymptotic expected value of $D/\phi = D^* \sim \chi_{n-p}^2$.

Definition 2.44. The Pearson X^2 estimator of the dispersion parameter is

$$\hat{\phi}_P = \frac{X^2}{n - p}$$

where X^2 is Pearson's statistic — this is based on the approximation $X^2/\phi \sim \chi_{n-p}^2$.

Remark 2.45. Where ϕ has been estimated from the data as $\hat{\phi}$, the additional sampling uncertainty means that strictly we should use the t distribution to form confidence intervals for entries of the model parameter β . For reasonable sizes, this ordinarily changes the confidence intervals by a negligible amount.

The Akaike Information Criteria (AIC)

Definition 2.46. The Akaike Information Criterion is

$$AIC = -2\ell(\hat{\beta}) + 2p.$$

A simple approach for choosing between two models is to prefer the one with the smaller AIC.

Remark 2.47. The AIC involves a trade-off between goodness-of-fit of the model and its complexity, as measured by the number of parameters.

2.7 Residuals

As with the normal linear models, we can use residuals to explore the fit of GLMs. For GLMs we require extensions of residual definitions to accommodate non-trivial mean-variance relationships.

Definition 2.48. For a single observation y_i , Pearson's residual is

$$r_{pi} = \frac{y_i - \mu_i}{\sqrt{V(\mu_i)}}.$$

Definition 2.49. For a single observation, y_i , the deviance residual is

$$r_{Di} = \text{sign}(y_i - \mu_i) \sqrt{d_i},$$

thus the deviance is $D = \sum_i r_{Di}^2$. Hence an observation's deviance residual is just the square root of its contribution to the deviance. Note that

$$\text{sign}(x) = \begin{cases} +1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Example 2.50. For the Poisson distribution, we have

Pearson's residual: $r_p = \frac{y_i - \mu_i}{\sqrt{\mu_i}}.$

Deviance residual: $r_D = \text{sign}(y_i - \mu_i) \{2(y_i \log(y_i / \mu_i) - y_i + \mu_i)\}^{1/2}.$

Remark 2.51. Similar with the residuals for linear models, we can standardise to account for each observation's leverage. To this end, we require the corresponding hat matrix for GLMs:

$$P = W^{1/2} X (X^T W X)^{-1} X^T W^{1/2}.$$

The standardised Pearson's and deviance residuals are obtained by dividing by $\sqrt{(1 - P_{ii})\hat{\phi}}$.

Remark 2.52. These residuals are approximately normal in general. However, the deviance residuals are the closest, for n large and n_i (for Binomial) large. In practice, one should not expect them to lie on a straight line in a QQ plot without first simulating data with roughly comparable sample size and parameter values.

Applied Statistical Inference

Section 3

Linear Mixed Models

3.1 Motivation

In all of the models considered so far, the observations have all been assumed to be independent random variables. Many statistical applications generate observations that are correlated, for example clinical trials that collect multiple measurements on each subject. Moreover, there are many settings where the quantity we are aiming to measure is the extent of correlation between observations.

Example 3.1. *A single random factor - heritability*

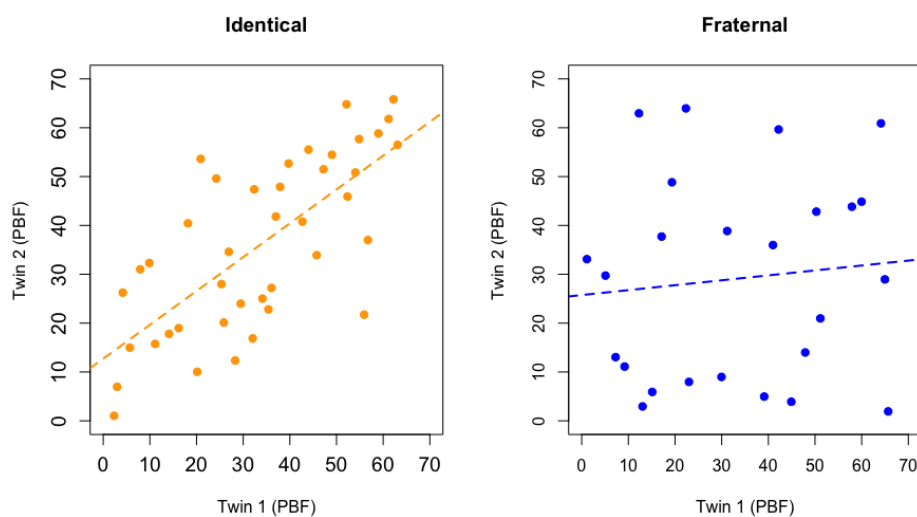


Figure 3.1: Percentage body fat (PBF) for identical and fraternal twins. More of the variability is shared between identical twins, giving a higher correlation. Data from [Faith et al. \(1999\)](#) via [Graham Coop's Genetics notes](#).

In agriculture, it is of interest to breed plants or animals selectively, to increase the yield or quality of the resulting food product. It is therefore important to understand how heritable a given trait is: a breeding programme can only be successful if the variability in a trait has a strong genetic component.

Consider an experiment in which k offspring from each of m hens are measured for a trait: this could be e.g. weight, or time to reach maturity. We would like to understand how heritable the trait is: if most of the variability is family-to-family heritability, then this suggests that a breeding programme can be successful, but if family-to-family variability is swamped by variability from other sources, such as the environment, then breeding will not be effective.

Here is a simple model for this set-up. There are two random contributions to the value of the trait for hen i from family j :

$$y_{ij} = \beta_0 + v_j + \epsilon_{ij},$$

where β_0 is the overall population mean of the trait, $v_j \sim N(0, \sigma_v^2)$ represents the genetic component of the trait common to all of the individuals in the j th family, and $\epsilon_{ij} \sim N(0, \sigma^2)$ corresponds to environmental and other uncontrolled effects.

The quantity of interest here is the intra-class correlation coefficient,

$$\rho = \frac{\sigma_v^2}{\sigma_v^2 + \sigma^2}.$$

Here, this gives is a measure of the extent to which a trait is heritable. We can of course also look at human traits in this way. 3.1 shows percentage body fat for identical and fraternal twins. Clearly the correlation between pairs of identical twins is substantially larger than for fraternal twins.

Example 3.2. *A random slopes model*

In Figure 3.2, we see structured data representing the relationship between socio-economic status and maths score for students from many different schools. In modelling such data, clearly it is inappropriate to ignore school entirely: the relationship between the two variables would plausibly be different from school to school. But equally, it is not optimal to treat different schools as entirely unrelated to each other, which is what we would be doing if we estimated a linear model separately for each school. This would mean estimating a very large number of parameters.

An intermediate proposal might be the following model,

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + v_j + \xi_j x_j + \epsilon_{ij},$$

where β_0 and β_1 are overall intercept and slope, respectively, and the random variables $v_j \sim N(0, \sigma_v^2)$, $\xi_j \sim N(0, \sigma_\xi^2)$ are the random intercept and slope, respectively, for each school, and $\epsilon_{ij} \sim N(0, \sigma^2)$ is random error. Instead of estimating two parameters for each dataset, we instead treat the parameters as random, and estimate the mean and variance of their distributions.

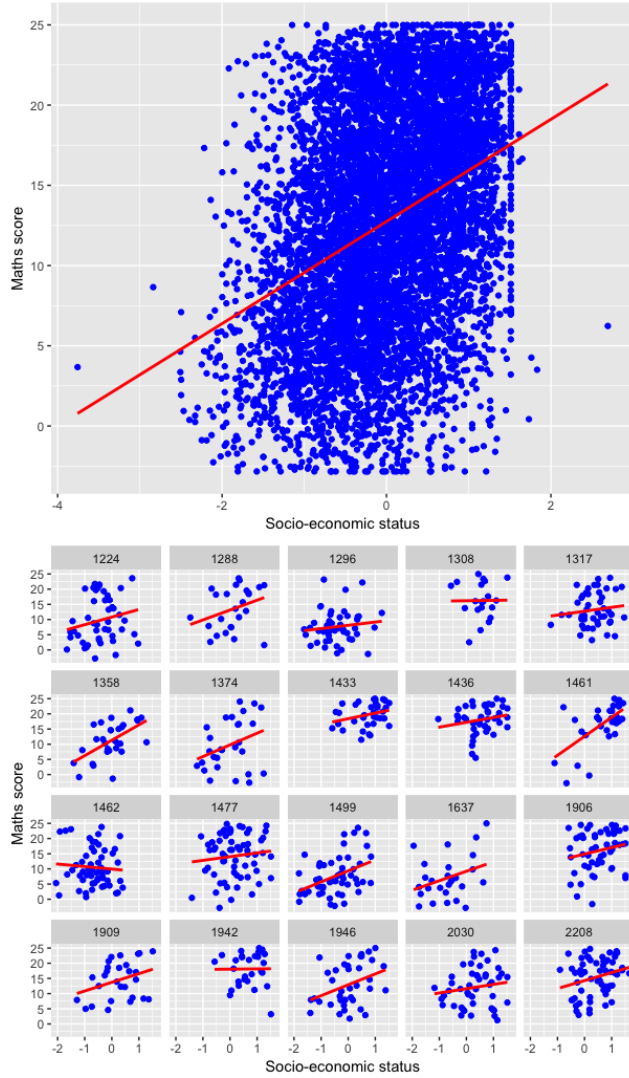


Figure 3.2: The relationship between maths score and socioeconomic ignoring their school (above) and with a model fit separately by school (below). Data taken from the [High School and Beyond study](#) .

Example 3.3. The Neyman-Scott problem

In Example 3.1, we could have treated the 'shared' component v_j as a fixed parameter, rather than as a random variable. What would happen in this case?

We take independent random variables $Y_{ij} \sim N(v_j, \sigma^2)$, for $i = 1, 2$ and $j = 1, 2, \dots, m$. In this case the maximum likelihood estimator of the mean

is just the sample average for each pair $\hat{v}_j = \frac{1}{2} (Y_{1j} + Y_{2j})$.

The maximum likelihood estimator for σ^2 is then (exercise)

$$\frac{1}{m} \sum_{j=1}^m S_j^2 = \frac{1}{m} \sum_{j=1}^m \frac{1}{2} \left((Y_{1j} - \hat{v}_j)^2 + (Y_{2j} - \hat{v}_j)^2 \right).$$

You can check that

$$S_j^2 = (Y_{1j} - \hat{v}_j)^2 + (Y_{2j} - \hat{v}_j)^2 = \frac{(Y_{1j} - Y_{2j})^2}{4},$$

but clearly $Y_{1j} - Y_{2j} \sim N(0, 2\sigma^2)$, so that $E[S_j^2] = \frac{1}{2}\sigma^2$.

This show that the maximum likelihood estimator of the variance σ^2 is biased.

Remark 3.4. It should be familiar to you from earlier work that maximum likelihood estimators of variance parameters can be biased: this happens even in the simple example $X_1, \dots, X_n \sim N(\mu, \sigma^2)$:

$$E[\hat{\sigma}^2] = \frac{n-1}{n}\sigma^2.$$

But in that case, the MLE is at least asymptotically unbiased: the bias goes to zero as $n \rightarrow \infty$. By contrast, in Example 3.3, the bias does not go to zero.

3.2 Model specification

Definition 3.5. Suppose that we have n observations overall from m subjects. The number of observations from each subject need not be the same: we suppose the j th subject contributes n_j observations, so that $\sum_{j=1}^m n_j = n$. A linear mixed model can be written

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{v} + \boldsymbol{\epsilon}, \quad \mathbf{v} \sim N(\mathbf{0}, \sigma_v^2 \mathbf{I}_m), \quad \boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma_e^2 \mathbf{I}_n),$$

where the vectors $\mathbf{v}$ and $\boldsymbol{\epsilon}$ are independent.

Definition 3.6. The matrix $\mathbf{Z}$ in definition 3.5 is called the model matrix.

Remark 3.7. We suppose an individual observation corresponds to precisely one group, so the model matrix $\mathbf{Z}$ is responsible for specifying which v_j corresponds to which observation - it has entries 0 or 1.

Definition 3.8. The variance components are the vector $\boldsymbol{\tau} = (\sigma_v^2, \sigma_e^2)$.

Remark 3.9. To convey the ideas cleanly, the linear mixed model defined above has only a single additional variance component. In principle we can

easily extend the ideas that follow to models such as Example 3.2, in which there were two group-level variance components (intercept and slope), as well as individual-level random error.

Example 3.10. *Simple example - intercept-only model*

Suppose $k = 2$ observations are made on each of $m = 3$ subjects so that $n = 3 \times 2 = 6$. For each observation $i = 1, 2$ on subjects $j = 1, 2, 3$, we have

$$y_{ij} = \beta_0 + v_j + \epsilon_{ij}, \quad v_j \sim N(0, \sigma_v^2), \quad \epsilon_{ij} \sim N(0, \sigma^2).$$

The model can then be written in vector notation as

$$\begin{pmatrix} y_{11} \\ y_{21} \\ y_{12} \\ y_{22} \\ y_{13} \\ y_{23} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \beta_0 + \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} + \begin{pmatrix} \epsilon_{11} \\ \epsilon_{21} \\ \epsilon_{12} \\ \epsilon_{22} \\ \epsilon_{13} \\ \epsilon_{23} \end{pmatrix}.$$

The distribution of the response

As a linear combination of multivariate normal variables, $\mathbf{Y}$ must also be multivariate normal. For a multivariate normal vector, the full distribution is determined by the mean vector and variance-covariance matrix. Using linearity of expectation,

$$\mathbf{E}[\mathbf{Y}] = \mathbf{E}[\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{v} + \boldsymbol{\epsilon}] = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{E}[\mathbf{v}] + \mathbf{E}[\boldsymbol{\epsilon}] = \mathbf{X}\boldsymbol{\beta} + \mathbf{0} + \mathbf{0} = \mathbf{X}\boldsymbol{\beta}.$$

Using the independence of $\mathbf{v}$ and $\boldsymbol{\epsilon}$, and the bilinearity of variance,

$$\text{Var}[\mathbf{Y}] = \text{Var}[\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{v} + \boldsymbol{\epsilon}] = \text{Var}[\mathbf{Z}\mathbf{v}] + \text{Var}[\boldsymbol{\epsilon}] = \mathbf{Z}\text{Var}[\mathbf{v}]\mathbf{Z}^T + \sigma^2\mathbf{I}_n = \mathbf{Z}\mathbf{Z}^T\sigma_v^2 + \sigma^2\mathbf{I}_n.$$

Definition 3.11. For the linear mixed model above, The intra-class correlation coefficient is

$$\rho = \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\epsilon^2}.$$

3.3 Estimation

Maximum Likelihood

Recall that the normal linear mixed model can be written as

$$\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \underbrace{\sigma_\epsilon^2(\mathbf{I}_n + \mathbf{Z}\boldsymbol{\Psi}\mathbf{Z}^T)}_{=: \mathbf{V}}), \quad \text{where} \quad \boldsymbol{\Psi} = \frac{\sigma_v^2}{\sigma_\epsilon^2}\mathbf{I}_m.$$

If the relative variance factor Ψ is known, we can estimate β using generalised least squares as in Section 1.10. However, the estimation of the variance components is often the purpose of the analysis. One way to do this is maximum likelihood.

Taking $V = I_n + Z\Psi Z^T$, the distribution of $\mathbf{Y}$ is

$$\frac{1}{(2\pi)^{(n/2)} |\sigma_\epsilon^2 V|^{1/2}} \exp \left\{ -\frac{1}{2\sigma_\epsilon^2} (\mathbf{y} - X\beta)^T V^{-1} (\mathbf{y} - X\beta) \right\}.$$

Thus the log-likelihood takes the form

$$\ell(\beta, \tau; \mathbf{y}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\sigma_\epsilon^2 V| - \frac{1}{2\sigma_\epsilon^2} (\mathbf{y} - X\beta)^T V^{-1} (\mathbf{y} - X\beta), \quad (3.1)$$

where $V = V(\Psi)$ depends on the variance components τ only through the relative variance factor Ψ .

We can then proceed to optimise to find the maximum likelihood estimates of β and τ using a numerical method, penalized least squares, for which efficient implementations exist. The details are beyond our scope, but are presented clearly in the package vignette for *lme4* by Bates et al.

Remark 3.12. The important constraint to respect is that the matrix V is positive definite, and so the method proceeds by estimating its Cholesky factor. One difficulty where there are several variance components, without overwhelming amounts of data, is that the maximum likelihood estimate of some variance components may be zero.

Restricted Maximum Likelihood - Motivation

Example 3.13. (Neyman-Scott continued)

We return to the Neyman-Scott problem from Example 3.3. There, we saw that the MLE approach can lead to a biased estimator of the variance. In contrast to our usual experience, the fixed effects, the μ_j , are considered nuisance parameters and the main interest is in estimating the variance σ^2 . We now discuss a method that can be used to estimate the parameters of interest without estimating the nuisance parameters.

Consider the transformation $B_j = Y_{1j} - Y_{2j}$. Then $B_1, \dots, B_m$ are independent and $N(0, 2\sigma^2)$ distributed. Notice that this transformation leads to a distribution that does not involve the nuisance parameters $\mu_1, \dots, \mu_m$.

The joint pdf of $B_1, \dots, B_m$ is then

$$f(b_1, \dots, b_m) = \prod_{j=1}^m \frac{1}{\sqrt{2\pi(2\sigma^2)}} \exp \left\{ -\frac{1}{2(2\sigma^2)} b_j^2 \right\}.$$

Hence the log-likelihood is

$$-\frac{m}{2} \log(4\pi\sigma^2) - \frac{1}{4\sigma^2} \sum_{j=1}^m B_j^2,$$

and it follows that

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{m}{2\sigma^2} + \frac{1}{4(\sigma^2)^2} \sum_{j=1}^m B_j^2.$$

Therefore, the MLE of σ^2 is

$$\hat{\sigma}^2 = \frac{1}{2m} \sum_{j=1}^m B_j^2.$$

Finally, we can compute the expectation of this MLE and find that it is indeed an unbiased estimator of σ^2 :

$$E(\hat{\sigma}^2) = \frac{1}{2m} \cdot m \cdot 2\sigma^2 = \sigma^2.$$

Remark 3.14. The idea in this example was to apply a transformation to the data that eliminates the fixed effects; we then use the transformed data to estimate the variance components. This idea can be applied much more generally.

Restricted Maximum Likelihood in general

Consider a linear model as in definition 3.5, such that X has full rank i.e. $\text{rank}(X) = p$ (with no real loss of generality). Let $L \in \mathbf{R}^{n \times (n-p)}$ be such that $\text{rank}(L) = n - p$ and

$$L^T X = 0.$$

Then multiplying the normal linear mixed model equation by L^T on the left we get

$$\begin{aligned} \underbrace{L^T \mathbf{Y}}_{=: \mathbf{w}} &= \underbrace{L^T X \boldsymbol{\beta}}_{=0} + L^T Z \mathbf{v} + L^T \boldsymbol{\epsilon} \\ \mathbf{w} &= L^T Z \mathbf{v} + L^T \boldsymbol{\epsilon}. \end{aligned}$$

After the transformation we have

$$\mathbf{w} \sim N(0, \sigma_\epsilon^2 L^T V L).$$

It follows that the joint pdf of $\mathbf{w}$ is

$$\frac{1}{(2\pi)^{(n-p)/2} |\sigma_\epsilon^2 L^T V L|^{1/2}} \exp \left\{ -\frac{1}{2\sigma_\epsilon^2} \mathbf{w}^T (L^T V L)^{-1} \mathbf{w} \right\}$$

so that the log-likelihood is

$$\ell_R(\boldsymbol{\tau}; \mathbf{w}) = -\frac{(n-p)}{2} \log(2\pi) - \frac{1}{2} \log |\sigma_\epsilon^2 L^T V L| - \frac{1}{2\sigma_\epsilon^2} \mathbf{w}^T (L^T V L)^{-1} \mathbf{w}.$$

Remark 3.15. To emphasise this is the restricted log-likelihood we use the subscript R .

Definition 3.16. The REML (*restricted maximum likelihood*) estimator of τ is defined as the maximiser of ℓ_R . As with ordinary MLE; such a maximiser satisfies the REML equation

$$\frac{\partial \ell_R}{\partial \tau} = \mathbf{0}.$$

Remark 3.17. Again, we rely on numerical methods to find the maximiser of the REML equation. After the random effects are estimated; the fixed effects are subsequently estimated as the maximizers of the likelihood evaluated where the variance components have their REML-estimated values.

Remark 3.18. We have already seen how such a matrix L can be constructed. Let

$$L^T = I_n - X(X^T X)^{-1} X^T,$$

then

$$\begin{aligned} L^T \mathbf{Y} &= L^T X \boldsymbol{\beta} + L^T Z \mathbf{v} + L^T \boldsymbol{\epsilon} \\ (I_n - X(X^T X)^{-1} X^T) \mathbf{Y} &= X \boldsymbol{\beta} - X(X^T X)^{-1} (X^T X) \boldsymbol{\beta} + L^T Z \mathbf{v} + L^T \boldsymbol{\epsilon} \\ \mathbf{Y} - X \hat{\boldsymbol{\beta}} &= L^T Z \mathbf{v} + L^T \boldsymbol{\epsilon}, \end{aligned}$$

which is independent of the fixed effects $\boldsymbol{\beta}$.

3.4 Inference

Fixed effects

To compare two nested models M_0 (smaller) and M_1 , we compute the difference of the log-likelihoods (or difference of deviances)

$$2 \left\{ \ell(\hat{\boldsymbol{\beta}}_1, \hat{\boldsymbol{\tau}}_1; \mathbf{y}) - \ell(\hat{\boldsymbol{\beta}}_0, \hat{\boldsymbol{\tau}}_0; \mathbf{y}) \right\},$$

where $\hat{\boldsymbol{\beta}}_0, \hat{\boldsymbol{\tau}}_0$ are the MLEs of the parameters for M_0 and $\hat{\boldsymbol{\beta}}_1, \hat{\boldsymbol{\tau}}_1$ are the MLEs of the parameters for M_1 .

Remark 3.19. If we want to use standard distributions to approximate the null distribution of our test statistic, we have two choices: either (wrongly) assume the variance parameters are equal to their estimated values, then perform a χ^2 -test, or try to take into account their uncertainty, then use an F -test. Alternatively, we can use a bootstrap approach.

Random effects

Testing whether to include the random effects or not involves the null hypothesis:

$$H_0 : \sigma_v^2 = 0.$$

The standard derivation of the asymptotic χ^2 distribution depends on the null hypothesis lying on the interior of the parameter space, which is not true here. Indeed the null distribution is known not to be χ^2 . A bootstrap approach usually works well in this context.

Remark 3.20. If the χ^2 approximation is used, then the test will tend to be conservative in the sense that the p -values will tend to be larger than they should be.

Remark 3.21. We cannot use the REML estimation method if we are comparing models with different fixed effect structures. The reason is that the REML estimates the random effects by eliminating the fixed effects (we transformed using L , which depends on the design matrix X). Restricted likelihoods for different models are not comparable. We use the maximum likelihood estimation method in this situation.

3.5 Prediction

Best prediction when all the parameters are known

When the fixed effects, β , and variance components, τ , are known, the best predictor for $\mathbf{v}$, in the sense of minimum mean squared error, is its conditional expectation given the data; that is,

$$\tilde{\mathbf{v}} := E(\mathbf{v}|\mathbf{y}) = \Psi Z^T V^{-1}(\mathbf{y} - X\beta).$$

To show $E(\mathbf{v}|\mathbf{y}) = \Psi Z^T V^{-1}(\mathbf{y} - X\beta)$, first consider the covariance between $\mathbf{Y}$ and $\mathbf{v}$:

$$E(\underbrace{(\mathbf{Y} - X\beta)}_{Z\mathbf{v} + \boldsymbol{\epsilon}} \mathbf{v}^T) = E(\underbrace{Z \mathbf{v} \mathbf{v}^T}_{\sigma_v^2 I_m}) + \underbrace{E(\boldsymbol{\epsilon} \mathbf{v}^T)}_{=0} = Z\sigma_v^2 \Psi = \sigma_v^2 Z\Psi.$$

The proof then follows by considering the joint over $\mathbf{Y}$ and $\mathbf{v}$ which is an $(n + m)$ -dimensional multivariate Gaussian distribution:

$$\begin{pmatrix} \mathbf{Y} \\ \mathbf{v} \end{pmatrix} \sim N \left(\begin{pmatrix} X\beta \\ 0 \end{pmatrix}, \sigma_v^2 \begin{pmatrix} I_n + Z\Psi Z^T & Z\Psi \\ \Psi Z^T & \Psi \end{pmatrix} \right)$$

where we also used that $\Psi^T = \Psi$. Now recall the result that for a multivariate normal vector written in terms of sub-vectors,

$$\begin{pmatrix} \mathbf{W}_a \\ \mathbf{W}_b \end{pmatrix} \sim N \left(\begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}, \begin{pmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{pmatrix} \right),$$

the conditional distribution can be written as

$$\mathbf{W}_b \mid \mathbf{W}_a = \mathbf{w}_a \sim N\left(\boldsymbol{\mu}_b + \Sigma_{ba}\Sigma_{aa}^{-1}(\mathbf{w}_a - \boldsymbol{\mu}_a), \Sigma_{bb} - \Sigma_{ba}\Sigma_{aa}^{-1}\Sigma_{ab}\right).$$

This allows us to obtain the conditional $\mathbf{v} \mid \mathbf{y}$ (i.e. the posterior) without performing an integration:

$$\tilde{\mathbf{v}} := E(\mathbf{v} \mid \mathbf{Y} = \mathbf{y}) = \Psi Z^T \underbrace{(I_n + Z\Psi Z^T)^{-1}}_{V^{-1}}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Once the best predictors $\tilde{\mathbf{v}}$ are obtained, the best predictor of $\eta = \mathbf{x}_*^T \boldsymbol{\beta} + \mathbf{z}_*^T \mathbf{v}$ is

$$\hat{\eta}_B := \mathbf{x}_*^T \boldsymbol{\beta} + \mathbf{z}_*^T \tilde{\mathbf{v}} = \mathbf{x}_*^T \boldsymbol{\beta} + \mathbf{z}_*^T \Psi Z^T V^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

The subscript B refers to the best predictor.

Best linear unbiased prediction (BLUP)

If the fixed effects, $\boldsymbol{\beta}$, are unknown but the variance components, $\boldsymbol{\tau}$, are known, we replace $\boldsymbol{\beta}$ by its MLE $\hat{\boldsymbol{\beta}}$, which is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V}^{-1} \mathbf{Y}.$$

After replacing $\boldsymbol{\beta}$ by its MLE, the predictor for $\eta := \mathbf{x}_*^T \boldsymbol{\beta} + \mathbf{z}_*^T \mathbf{v}$ becomes

$$\hat{\eta}_{BLUP} = \mathbf{x}_*^T \hat{\boldsymbol{\beta}} + \mathbf{z}_*^T \tilde{\mathbf{v}}_{\hat{\boldsymbol{\beta}}} = \mathbf{x}_*^T \hat{\boldsymbol{\beta}} + \mathbf{z}_*^T \Psi Z^T V^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

This is the best linear unbiased predictor (BLUP) of η , in the sense that (i) it is linear in $\mathbf{y}$, (ii) its expected value is equal to η and (iii) it minimises the MSE among all linear unbiased predictors. The vector

$$\tilde{\mathbf{v}}_{\hat{\boldsymbol{\beta}}} := \Psi Z^T V^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$$

is also called the BLUP of $\mathbf{v}$.

Empirical BLUP

In practice, both the fixed effects and the variance components are typically unknown. In such cases, one replaces the vector of variance components, $\boldsymbol{\tau}$, which is involved in the expression of BLUP, by, $\hat{\boldsymbol{\tau}}$. That is, instead of predicting the random effects by $\tilde{\mathbf{v}}$ we use

$$\hat{\mathbf{v}} := E(\mathbf{v} \mid \mathbf{Y} = \mathbf{y}, \boldsymbol{\tau} = \hat{\boldsymbol{\tau}}) = \Psi Z^T V_{\hat{\boldsymbol{\tau}}}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$$

where $V_{\hat{\boldsymbol{\tau}}}$ is V with an estimator of $\boldsymbol{\tau}$ plugged in. The resulting predictor, often called empirical BLUP or eBLUP, is

$$\hat{\eta}_{eBLUP} = \mathbf{x}_*^T \hat{\boldsymbol{\beta}} + \mathbf{z}_*^T \hat{\mathbf{v}} = \mathbf{x}_*^T \hat{\boldsymbol{\beta}} + \mathbf{z}_*^T \Psi Z^T V_{\hat{\boldsymbol{\tau}}}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

Remark 3.22. In this context, the random effects, $\mathbf{v}$, are not considered as fixed parameters: they are **random variables**. So instead of estimating their values, we **predict** them using their conditional expectations.

3.6 Inference Continued

Inference about the fixed effects, β

Let us return to the un-restricted log-likelihood

$$\ell(\beta, \tau; \mathbf{y}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\sigma_\epsilon^2 V_\tau| - \frac{1}{2\sigma_\epsilon^2} (\mathbf{y} - X\beta)^T V_\tau^{-1} (\mathbf{y} - X\beta). \quad (3.2)$$

For fixed τ , taking derivative of this log-likelihood with respect to β gives

$$\frac{\partial \ell}{\partial \beta} = X^T V_\tau^{-1} (\mathbf{y} - X\beta)$$

so that the MLE of β is the solution of

$$X^T V_\tau^{-1} X \hat{\beta} = X^T V_\tau^{-1} \mathbf{y},$$

which we recognise. Recall that $V_\tau := (I_n + Z\Psi Z^T)$ where $\Psi = \frac{\sigma_v^2}{\sigma_\epsilon^2} I_m$ — therefore V_τ depends on the variance components $\tau = (\sigma_\epsilon^2, \sigma_v^2)$. The (profile) log-likelihood of τ is

$$\ell(\tau; \mathbf{y}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\sigma_\epsilon^2 V_\tau| - \frac{1}{2\sigma_\epsilon^2} (\mathbf{y} - X\hat{\beta})^T V_\tau^{-1} (\mathbf{y} - X\hat{\beta}).$$

The Fisher information of β is

$$\mathcal{J}(\beta) = X^T V_\tau^{-1} X.$$

Remark 3.23. In practice, the estimated value of $\hat{\tau}$ is plugged into the Fisher information, from which we can find the standard error for the MLE $\hat{\beta}$ in the form

$$\begin{aligned} \hat{\beta} &= \hat{\beta}_{\hat{\tau}} \\ \mathcal{J}(\hat{\beta}) &= X^T V_{\hat{\tau}}^{-1} X. \end{aligned}$$

Remark 3.24. Uncertainty in the linear model can be hard to account for using asymptotic results. It is common instead to use a bootstrap approach, which we will explore in practical tutorials.

